Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 270
Manufacturing Technology

Lecture 6: Material Removal Processes

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Lecture Objectives

- Understand material removal processes
- Explain turning, drilling and milling operations.

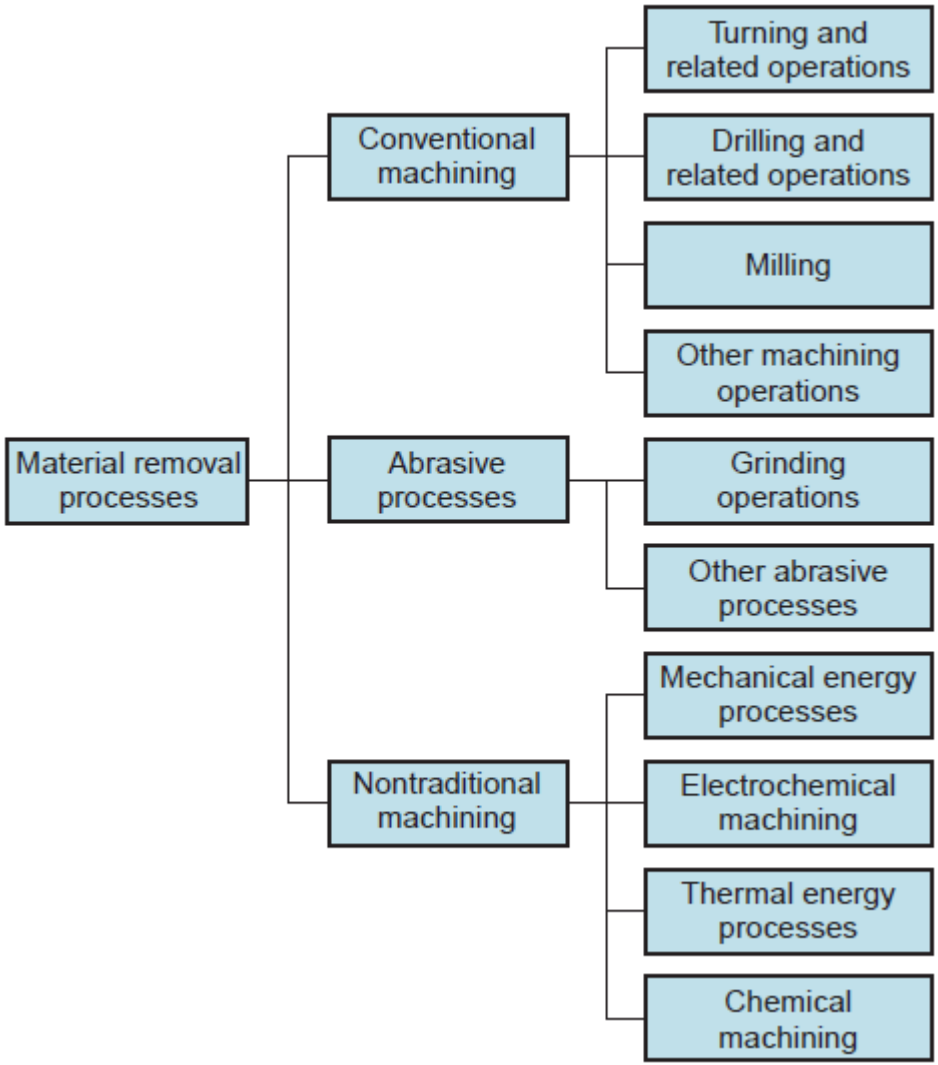
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Material Removal Processes

- The material removal processes are a family of shaping operations in which excess material is removed from a starting work-part so that what remains is the desired final geometry.
- The most important material removal process is conventional machining, in which a sharp cutting tool is used to mechanically cut the material to achieve the desired geometry.
- The three principal machining processes are **turning, drilling, and milling.**

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Classification of material removal processes



Machining

- Machining is a manufacturing process in which a sharp cutting tool is used to cut away material to leave the desired part shape.
- The predominant cutting action in machining involves shear deformation of the work material to form a chip; as the chip is removed, a new surface is exposed.
- Machining is most frequently applied to shape metals.

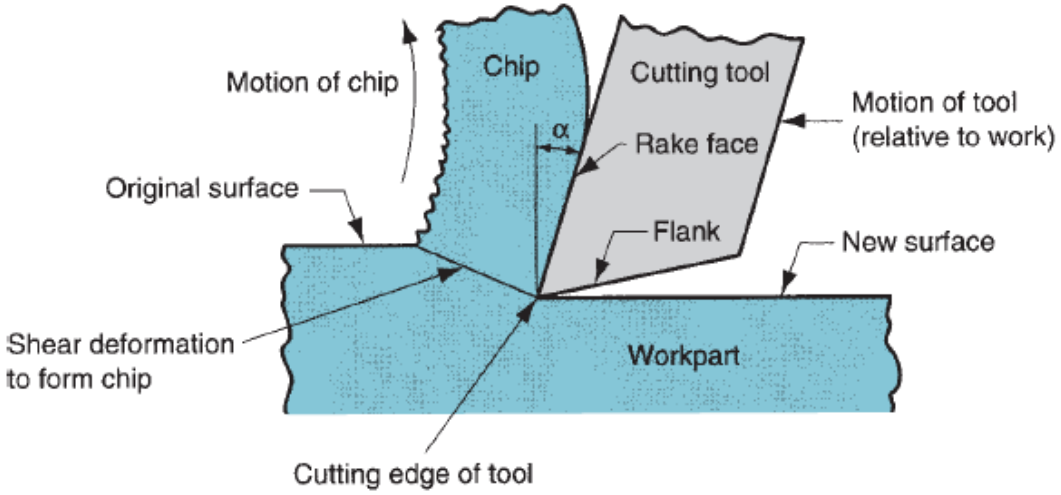


Fig. Cross-section of the machining process.

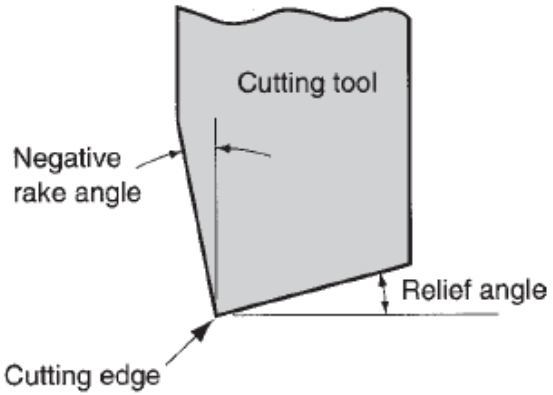


Fig. Tool with negative rake angle.

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Advantages of Machining

- **Variety of work materials**: Machining can be applied to a wide variety of work materials. Virtually all solid metals can be machined. Ceramics pose difficulties because of their high hardness and brittleness; however, most ceramics can be successfully cut by abrasive machining processes.

- **Variety of part shapes and geometric features**: Machining can be used to create any regular geometries, such as flat planes, round holes, and cylinders. By introducing variations in tool shapes and tool paths, irregular geometries can be created, such as screw threads and T-slots. By combining several machining operations in sequence, shapes of almost unlimited complexity and variety can be produced.

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Advantages of Machining

- **Dimensional accuracy**. Machining can produce dimensions to very close tolerances. Some machining processes can achieve tolerances of ±0.025 mm (±0.001 in), much more accurate than most other processes.
- **Good surface finishes**: Machining is capable of creating very smooth surface finishes. Roughness values less than 0.4 microns (16 μ-in.) can be achieved in conventional machining operations. Some abrasive processes can achieve even better finishes.

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Disadvantages of Machining

- **Waste of material**: Machining is inherently wasteful of material. The chips generated in a machining operation are wasted material. Although these chips can usually be recycled, they represent waste in terms of the unit operation.

- **Time consuming**: A machining operation generally takes more time to shape a given part than alternative shaping processes such as casting or forging.

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Machining Technology

- Machining is not just one process; it is a group of processes.
- The common feature is the use of a cutting tool to form a chip that is removed from the work-part.
- To perform the operation, relative motion is required between the tool and work.
- This relative motion is achieved in most machining operations by means of a primary motion, called the ***cutting speed***, and a secondary motion, called the ***feed***.
- The shape of the tool and its penetration into the work surface, combined with these motions, produces the desired geometry of the resulting work surface.
- Machining is generally performed after other manufacturing processes such as casting or bulk deformation (e.g., forging, bar drawing).
- The other processes create the general shape of the starting work-part, and machining provides the final geometry, dimensions, and finish.

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Types of Machining Operations

- The three most common types of machining operations are **turning, drilling, and milling.**
- In **turning**, a cutting tool with a single cutting edge is used to remove material from a rotating workpiece to generate a cylindrical shape.
- The ***speed*** motion in turning is provided by the rotating work-part, and the ***feed*** motion is achieved by the cutting tool moving slowly in a direction parallel to the axis of rotation of the workpiece.

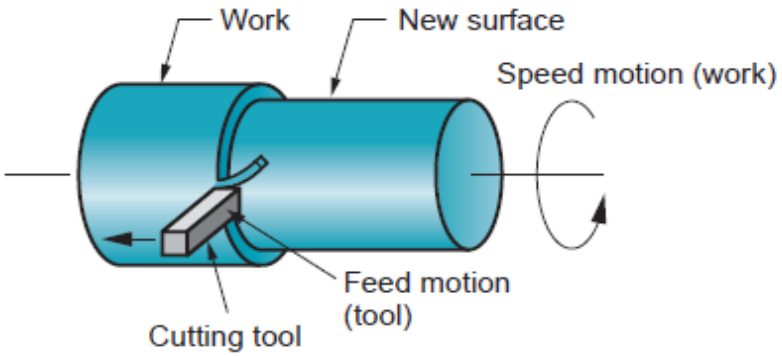
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Types of Machining Operations

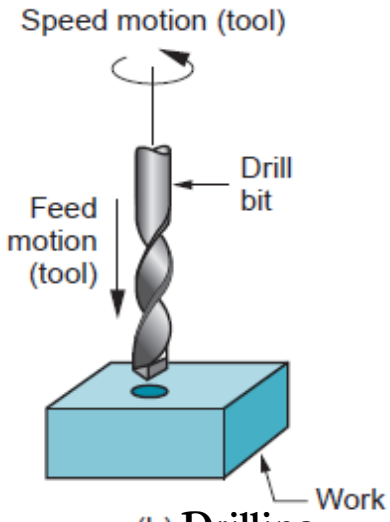
- **Drilling** is used to create a round hole.
- It is accomplished by a rotating tool that typically has two cutting edges.
- The tool is fed in a direction parallel to its axis of rotation into the workpart to form the round hole.
- In **milling**, a rotating tool with multiple cutting edges is fed slowly across the work material to generate a plane or straight surface.
- The direction of the feed motion is perpendicular to the tool's axis of rotation.
- The speed motion is provided by the rotating milling cutter.
- The two basic forms of milling are ***peripheral milling*** and ***face milling.***

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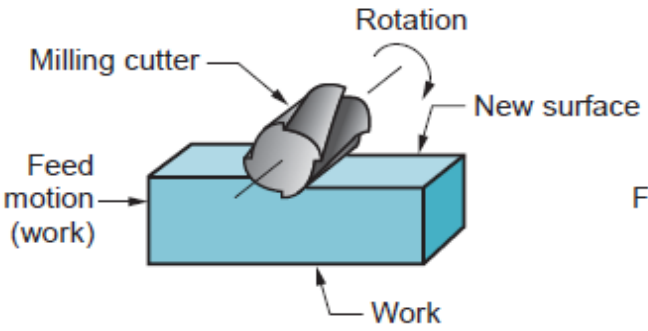
Types of Machining Operations



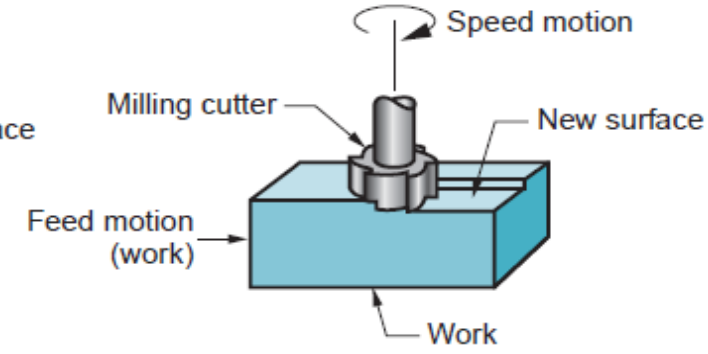
(a) **Turning**



(b) **Drilling**



(c) **Peripheral Milling**



(d) **Face Milling**

The Cutting Tool

- A cutting tool has one or more sharp cutting edges and is made of a material that is harder than the work material.
- The cutting edge serves to separate a chip from the parent work material.
- Connected to the cutting edge are two surfaces of the tool: **the rake face** and **the flank**.
- The rake face, which directs the flow of the newly formed chip, is oriented at a certain angle called the **rake angle α**.
- It is measured relative to a plane perpendicular to the work surface.
- The rake angle can be positive or negative.
- The flank of the tool provides a clearance between the tool and the newly generated work surface, thus protecting the surface from abrasion, which would degrade the finish.
- The flank surface is oriented at an angle called the **relief angle**.

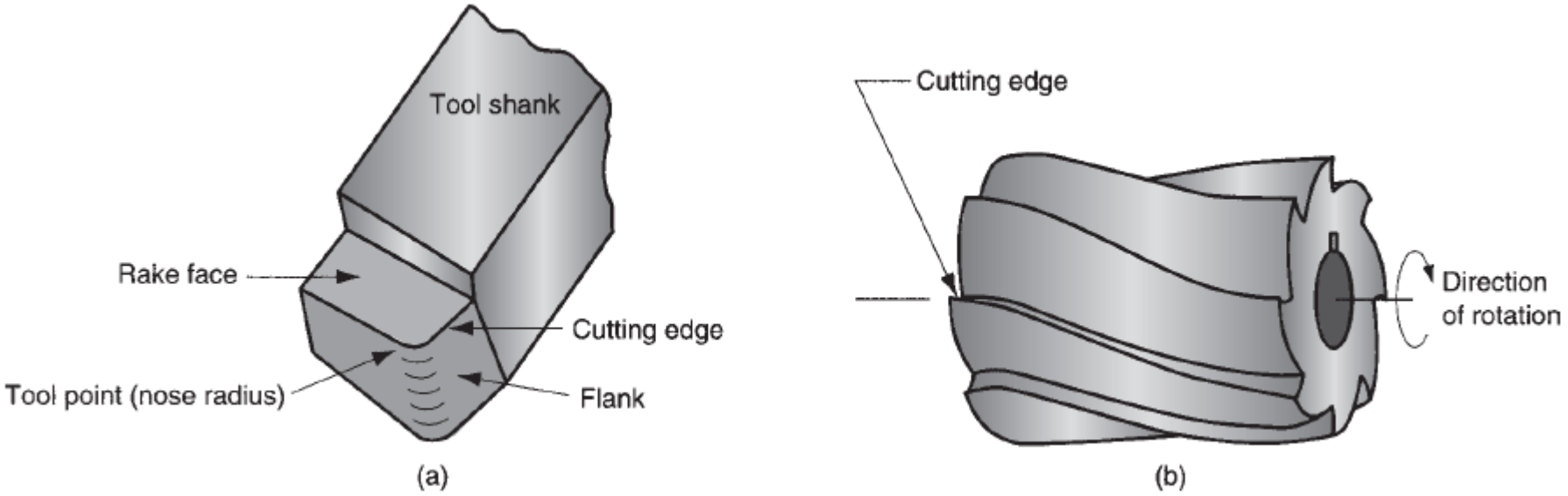
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The Cutting Tool

- There are two basic types of cutting tools: **single-point tools** and **multiple-cutting-edge tools.**
- A **single-point tool** has one cutting edge and is used for operations such as turning. During machining, the point of the tool penetrates below the original work surface of the part. The point is usually rounded to a certain radius, called the **nose radius.**
- **Multiple-cutting-edge tools** have more than one cutting edge and usually achieve their motion relative to the work-part by rotating.
- Drilling and milling use rotating multiple-cutting-edge tools.

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The Cutting Tool



(a) A single-point tool showing rake face, flank, and tool point
(b) A helical milling cutter, representative of tools with multiple cutting edges.

Cutting Conditions

- Relative motion is required between the tool and work to perform a machining operation.
- The primary motion is accomplished at a certain **cutting speed (v)**.
- In addition, the tool must be moved laterally across the work. This is a much slower motion, called **the feed (f)**.
- The remaining dimension of the cut is the penetration of the cutting tool below the original work surface, called **the depth of cut (d).**

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Cutting Conditions

- Collectively, ***speed, feed***, and ***depth of cut*** are called **the cutting conditions.**
- They form the three dimensions of the machining process, and can be used to calculate the material removal rate for certain processes.
- The material removal rate is given by:

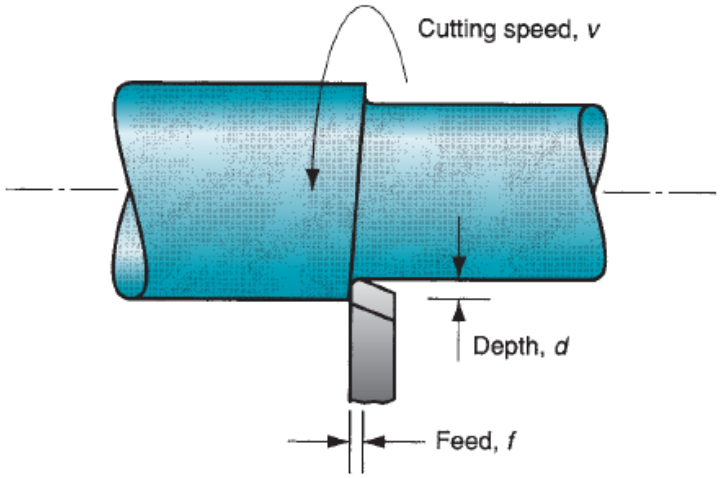
$$\boldsymbol{R_{MR} = vfd}$$

Where R_{MR} is material removal rate (m^3/s)

v is cutting speed (m/s)

f is feed (m)

d is depth of cut (mm)



Machining Operations

- Machining operations are usually divide into two categories: **roughing cuts** and **finishing cuts.**
- **Roughing cuts** are used to remove large amounts of material from the starting work-part as rapidly as possible, in order to produce a shape close to the desired form, but leaving some material on the piece for a subsequent finishing operation.
- **Finishing cuts** are used to complete the part and achieve the final dimensions, tolerances, and surface finish.
- Roughing operations are performed at high feeds and depths while finishing operations are carried out at low feeds and depths.
- Cutting speeds are lower in roughing than in finishing.

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Cutting Fluid

- A cutting fluid is often applied to the machining operation to cool and lubricate the cutting tool.
- Determining whether a cutting fluid should be used, and, if so, choosing the proper cutting fluid, is usually included within the scope of cutting conditions.
- Given the work material and tooling, the selection of these conditions is very influential in determining the success of a machining operation.

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Machine Tools

- A machine tool is used to ***hold the work-part, position the tool relative to the work, and provide power*** for the machining process at the speed, feed, and depth that have been set.
- By controlling the tool, work, and cutting conditions, machine tools permit parts to be made with great accuracy and repeatability, to tolerances of 0.025 mm and better.
- The term machine tool applies to any power-driven machine that performs a machining operation, including grinding.
- The term is also applied to machines that perform metal forming and press-working operations.

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Machine Tools

- The traditional machine tools used to perform turning, drilling, and milling are lathes, drill presses, and milling machines, respectively.
- Conventional machine tools are usually tended by a human operator, who loads and unloads the work-parts, changes cutting tools, and sets the cutting conditions.
- Many modern machine tools are designed to accomplish their operations with a form of automation called computer numerical control.

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Machining operations and machine tools

Learning objectives

- Describe turning and related operations
- Describe drilling and related Operations
- Explain milling operations
- Describe machining centers and turning centers
- Describe other machining operations
- Explain machining operations for special geometries
- Describe high speed machining processes

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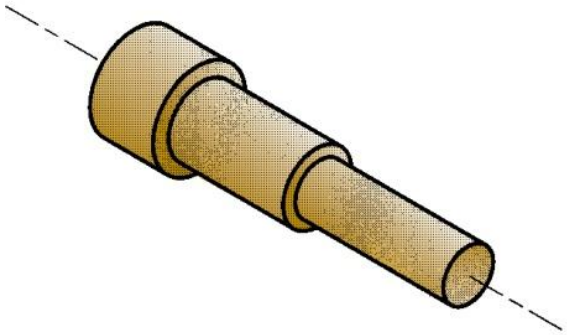
Machining

- Material removal process in which a sharp cutting tool is used to mechanically cut away material so that the desired part geometry remains.
- Most common application: metal parts
- Most versatile of all manufacturing processes for producing a variety of part shapes and geometric features with high precision and accuracy.

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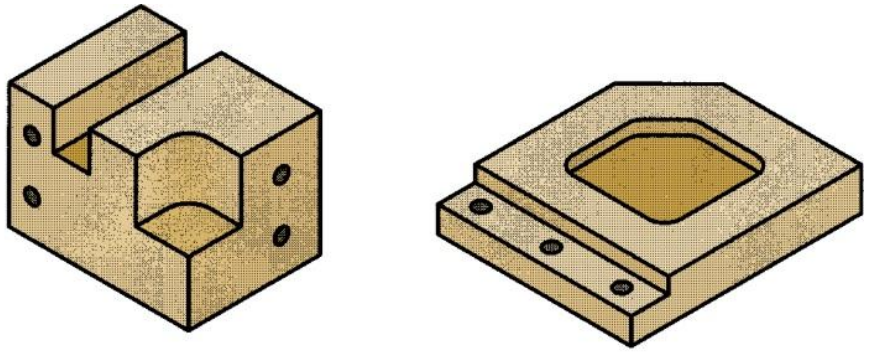
Classification of Machined Parts

- **Rotational parts** such as cylindrical or disk-like shapes
- **Non-rotational parts** such as block-like and plate-like shapes.



(a)

cylindrical or disk-like shape



(b)

block-like and plate-like

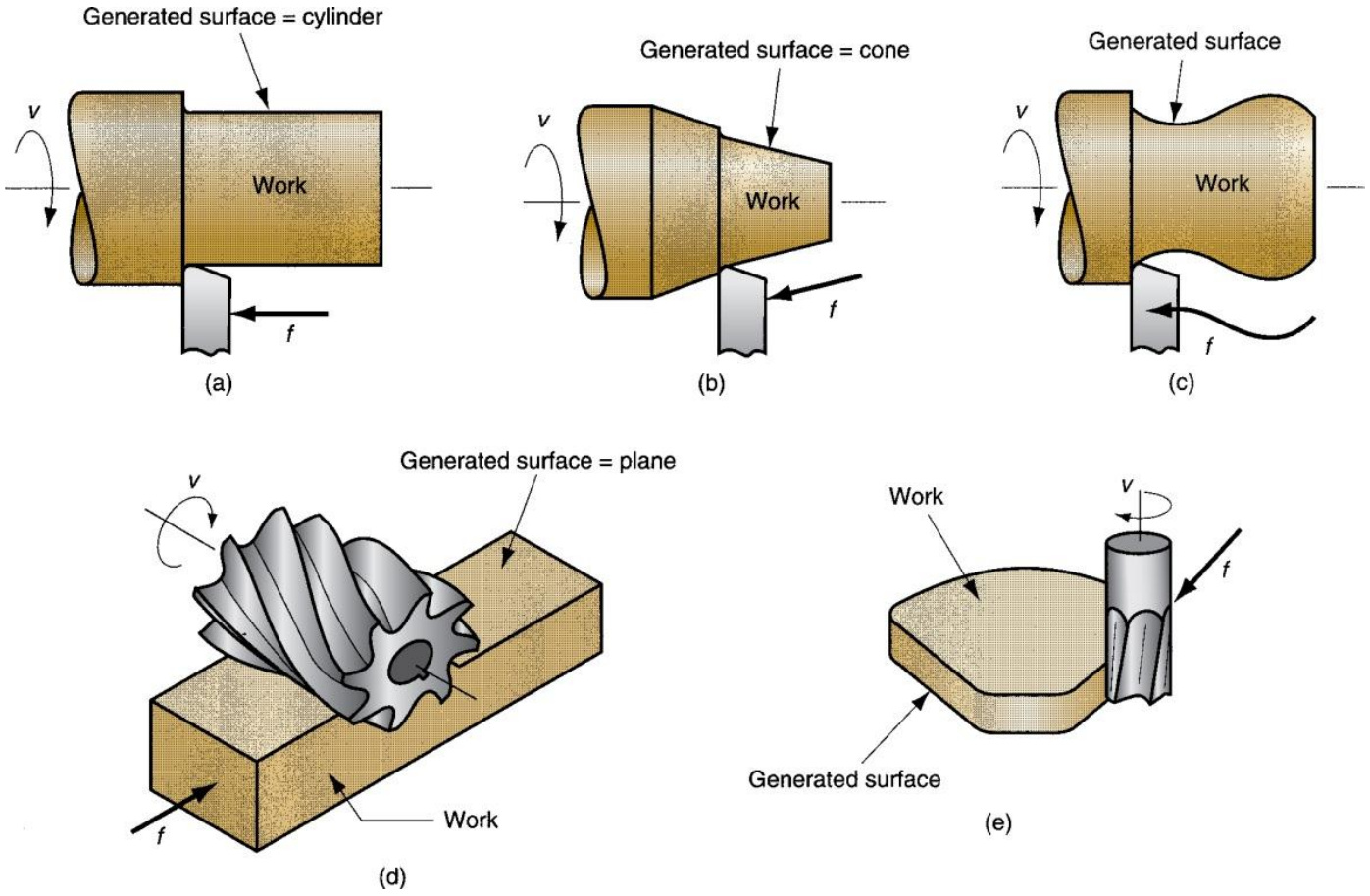
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Machining Operations and Part Geometry

- Each machining operation produces a characteristic part geometry due to two factors:
  - ✓ **Relative motions between tool and work-part**
    - ✓ ***Generating shape*** – part geometry determined by feed trajectory of cutting tool
  - ✓ **Shape of the cutting tool**
    - ✓ ***Forming shape*** – part geometry is created by the shape of the cutting tool

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Generating shape



(a) straight turning

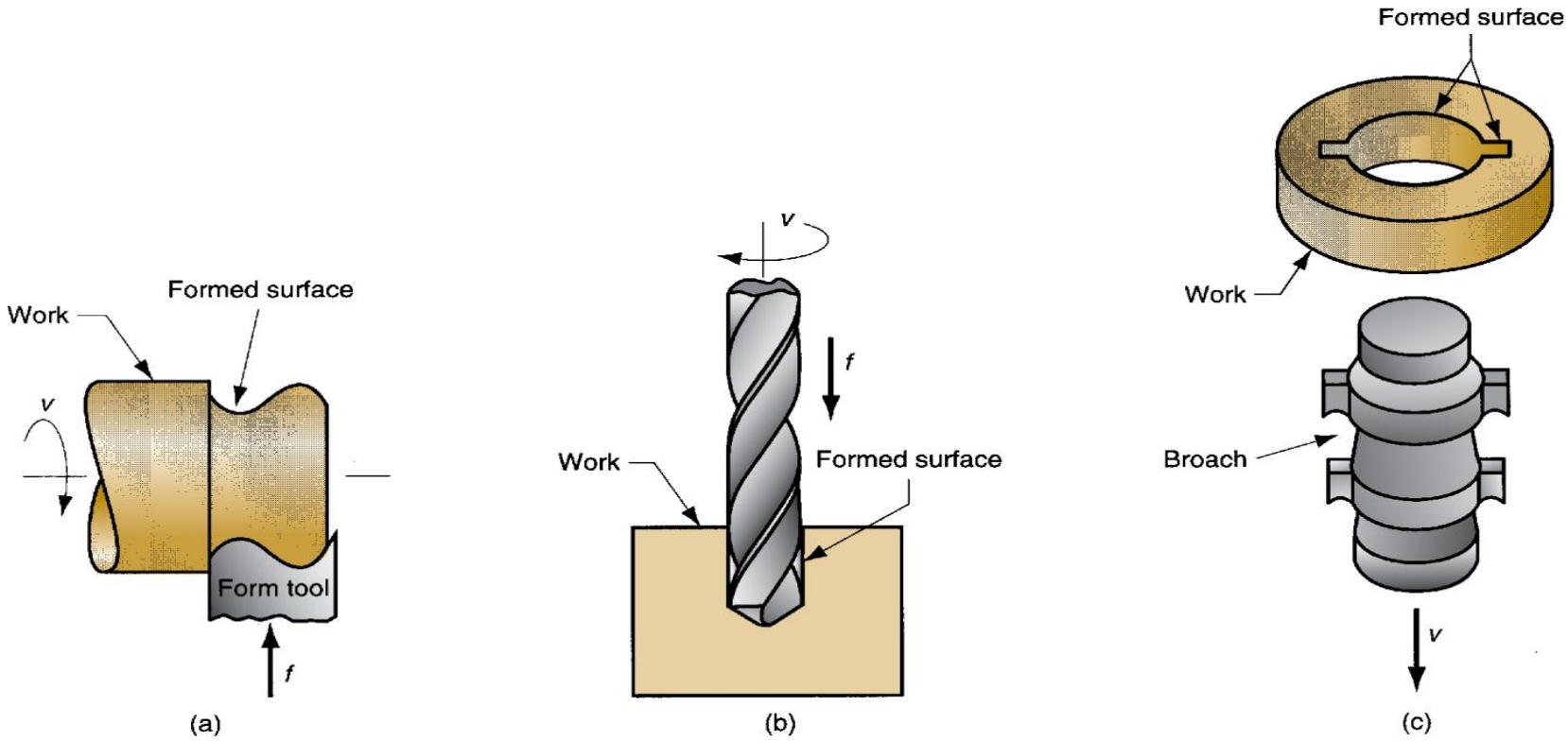
(b) taper turning

(c) contour turning

(d) plain milling

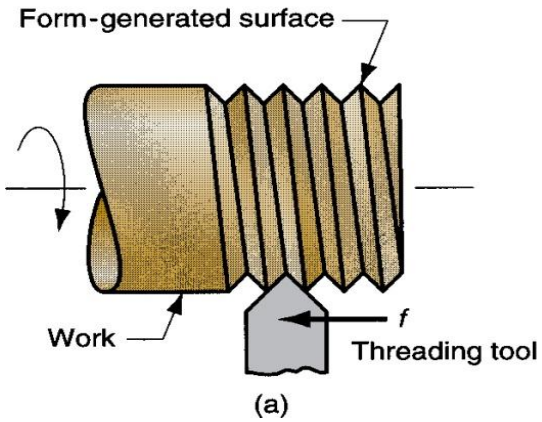
(e) profile milling

Forming to Create Shape

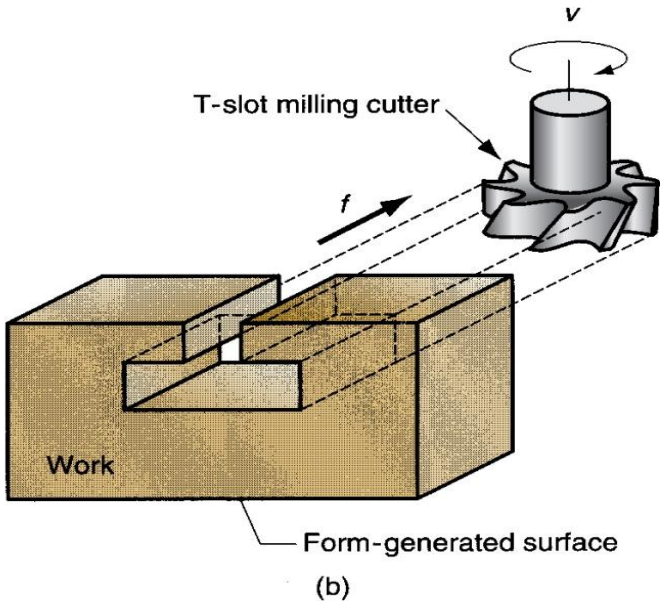


(a) form turning, (b) drilling, and (c) broaching

Forming and Generating



(a) thread cutting on a lathe



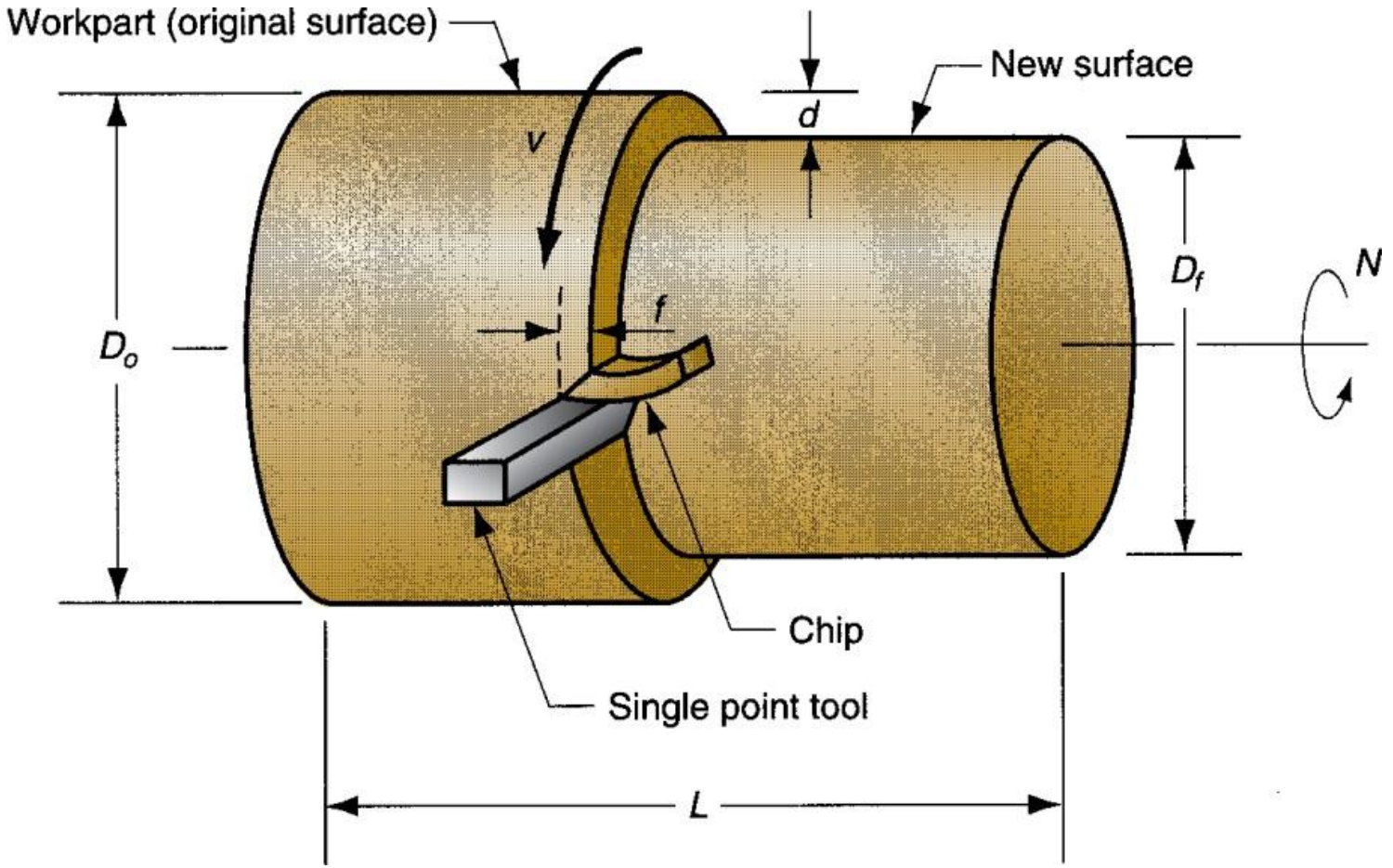
(b) slot milling

Turning

- Single point cutting tool removes material from a rotating work-piece to generate a cylindrical shape.
- Performed on a machine tool called a ***lathe.***
- Variations of turning performed on a lathe
  - ✓ Facing
  - ✓ Contour turning
  - ✓ Chamfering
  - ✓ Cutoff
  - ✓ Threading

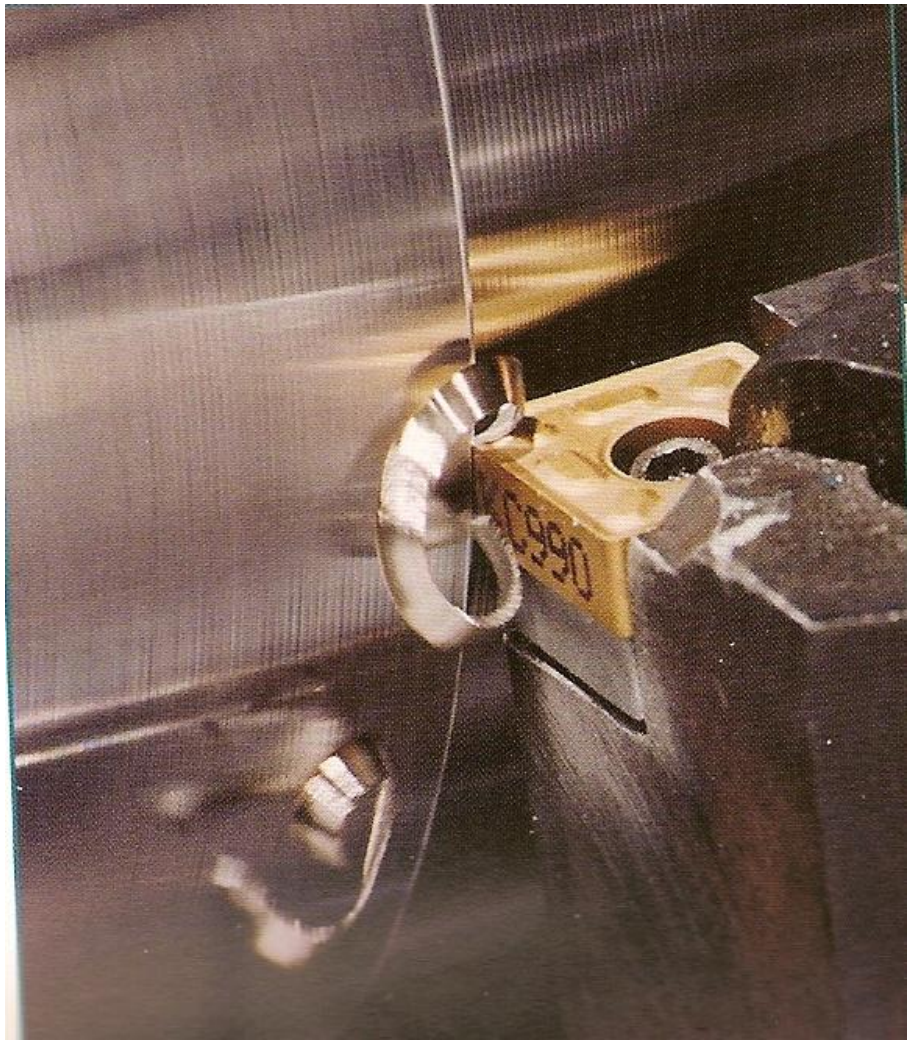
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Turning operation

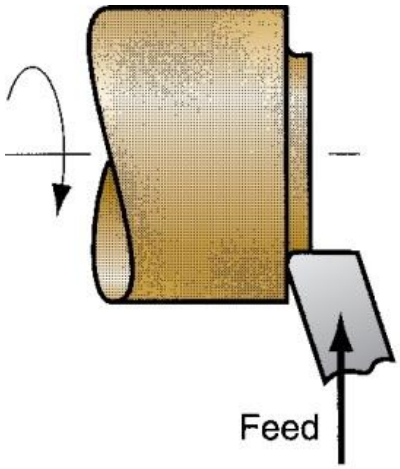


Turning operation

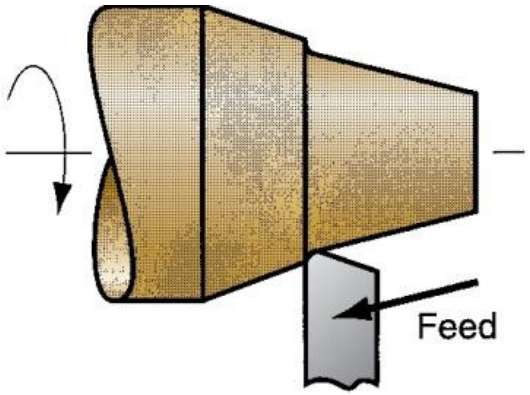
Fig. Close-up view of a turning operation on steel using a titanium nitride coated carbide cutting



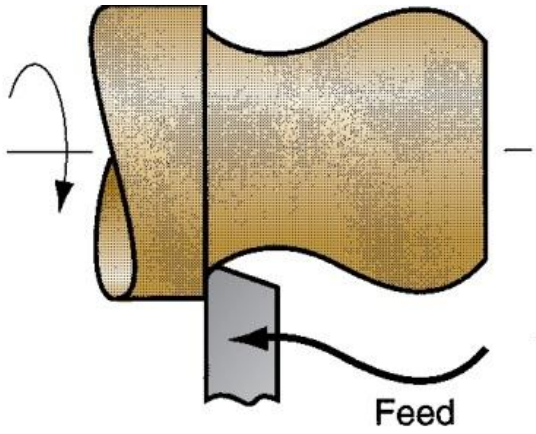
Operations related to turning



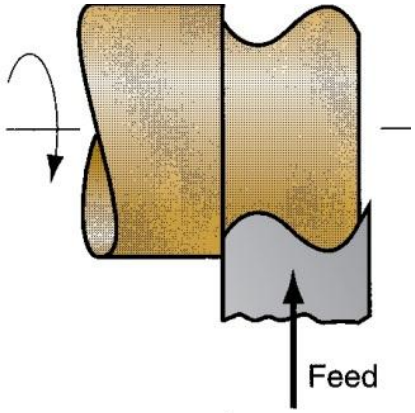
(a) Facing



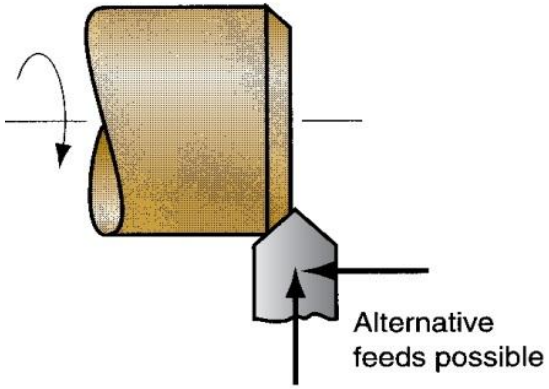
(b) Taper turning



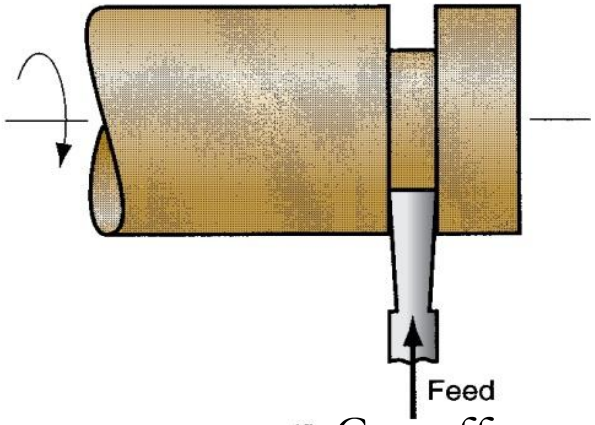
(c) Contour turning



(d) Form turning

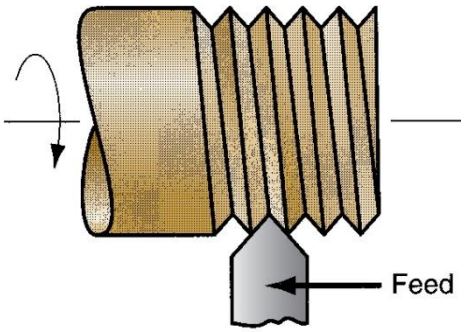


(e) Chamfering

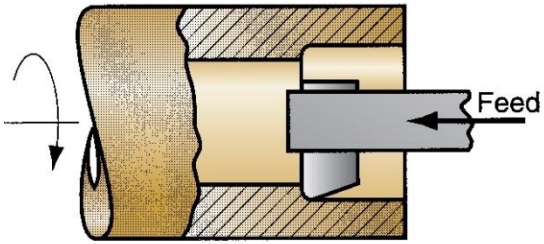


(f) Cut-off

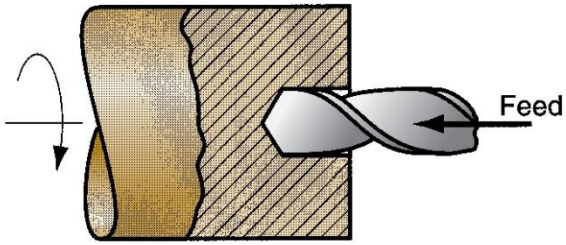
Operations related to turning



(g) Threading

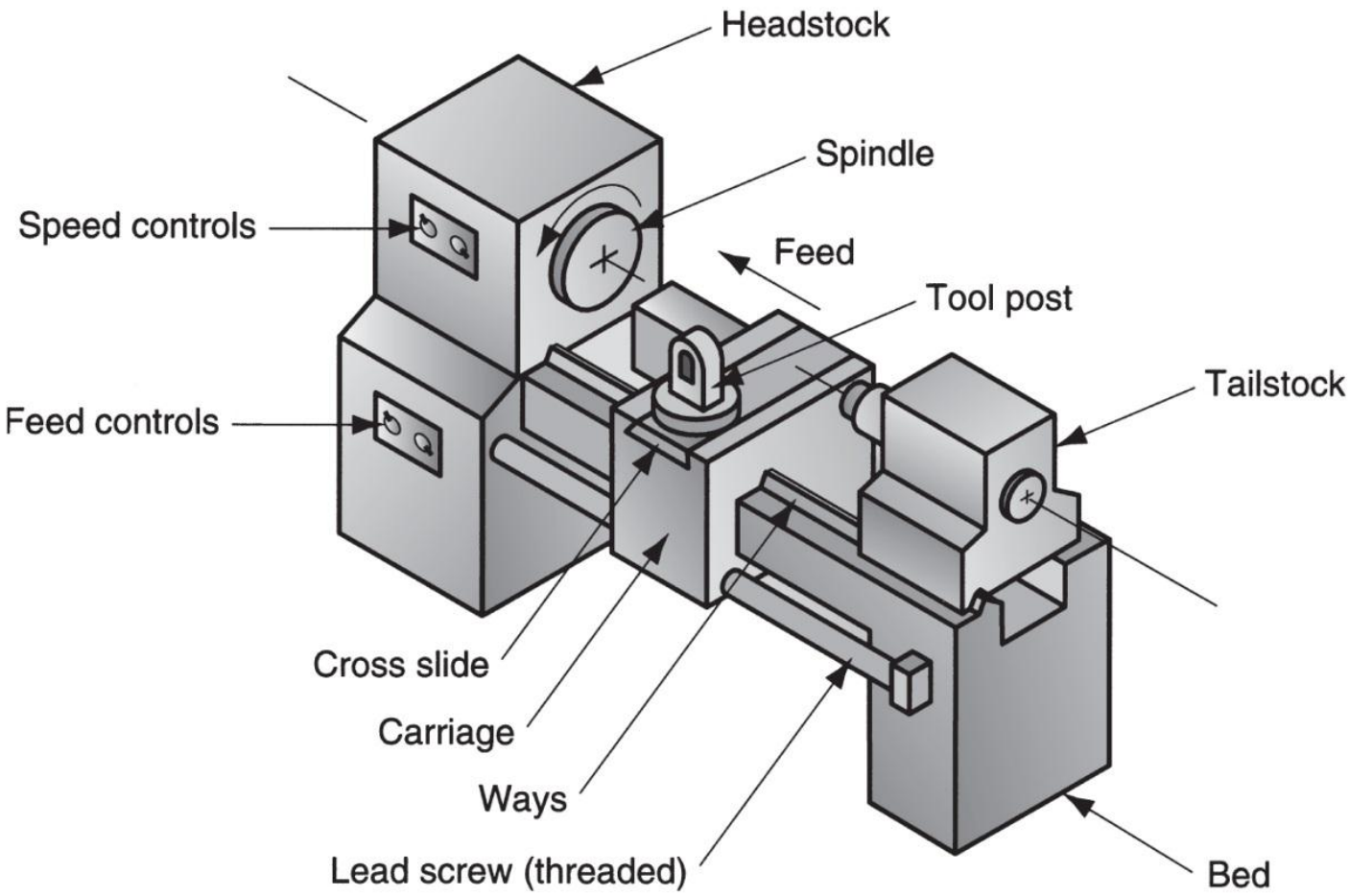


(h) boring



(i) drilling

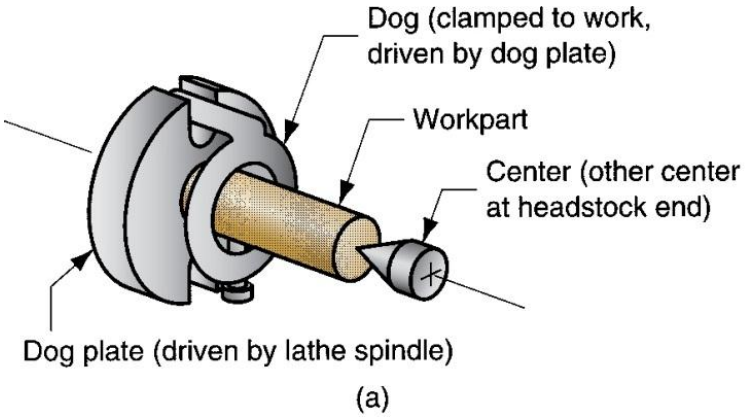
Engine Lathe



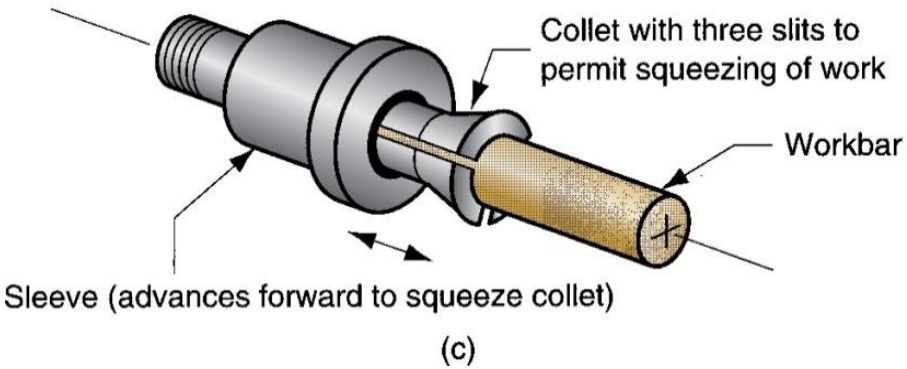
Operation of Lathe Machine



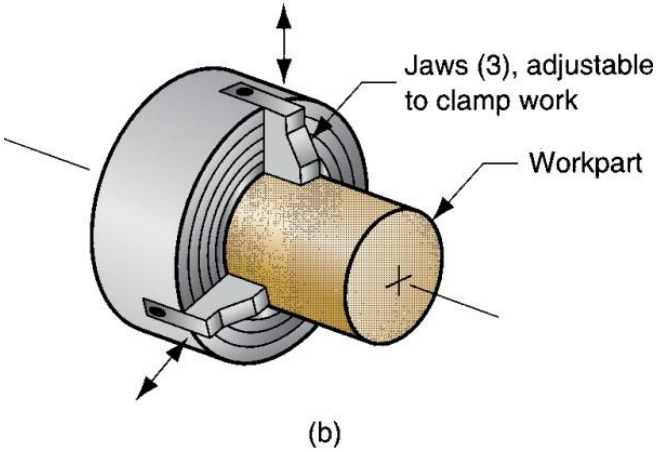
Methods of holding work-piece in a lathe



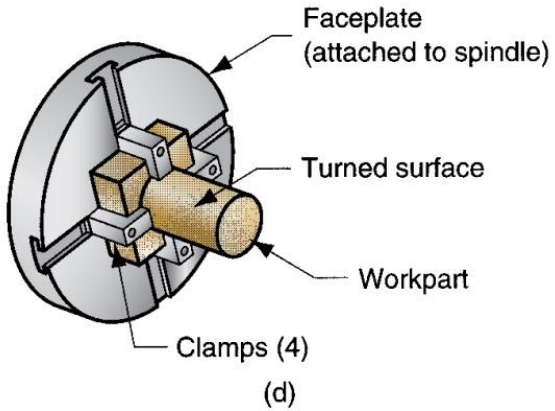
Holding the work between centers



Collet



Chuck



Face plate

Other Production Turning Machines

- Turret lathe
- Chucking machine
- Bar machine
- Automatic screw machine
- Multiple spindle bar machine

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Tureth Lathe

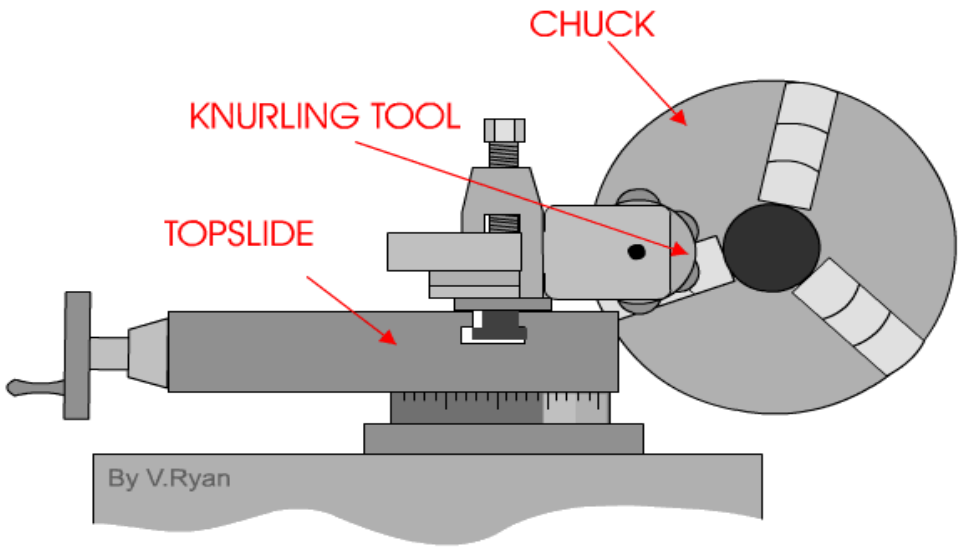
- Tailstock replaced by "turret" that holds up to six tools.
- Tools rapidly brought into action by indexing the turret.
- Tool post replaced by four-sided turret to index four tools.

Applications: high production work that requires a sequence of cuts on the part.



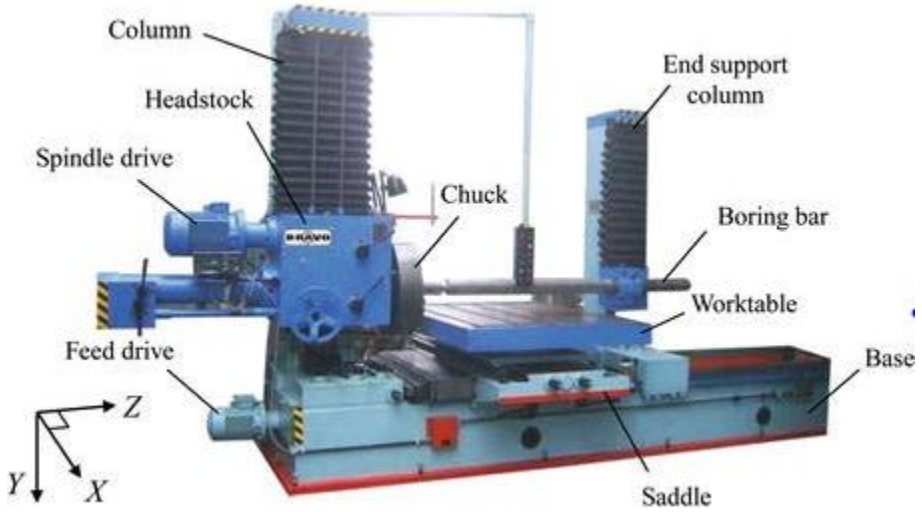
Chucking Machine

- Uses chuck in its spindle to hold work-part
- No tailstock, so parts cannot be mounted between centers
- Cutting tool actions controlled automatically
- Operator's job: to load and unload parts
- **Applications:** short, light-weight parts



Bar Machine

- Similar to chucking machine except collet replaces chuck, permitting long bar stock to be fed through headstock
  - ✓ At the end of the machining cycle, a cutoff operation separates the new part
- Highly automated
  - ✓ Computer numerical control
- **Applications**: high production of rotational parts



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Automatic Screw Machine

- Same as automatic bar machine but smaller.

Applications: high production of screws and similar small hardware items.

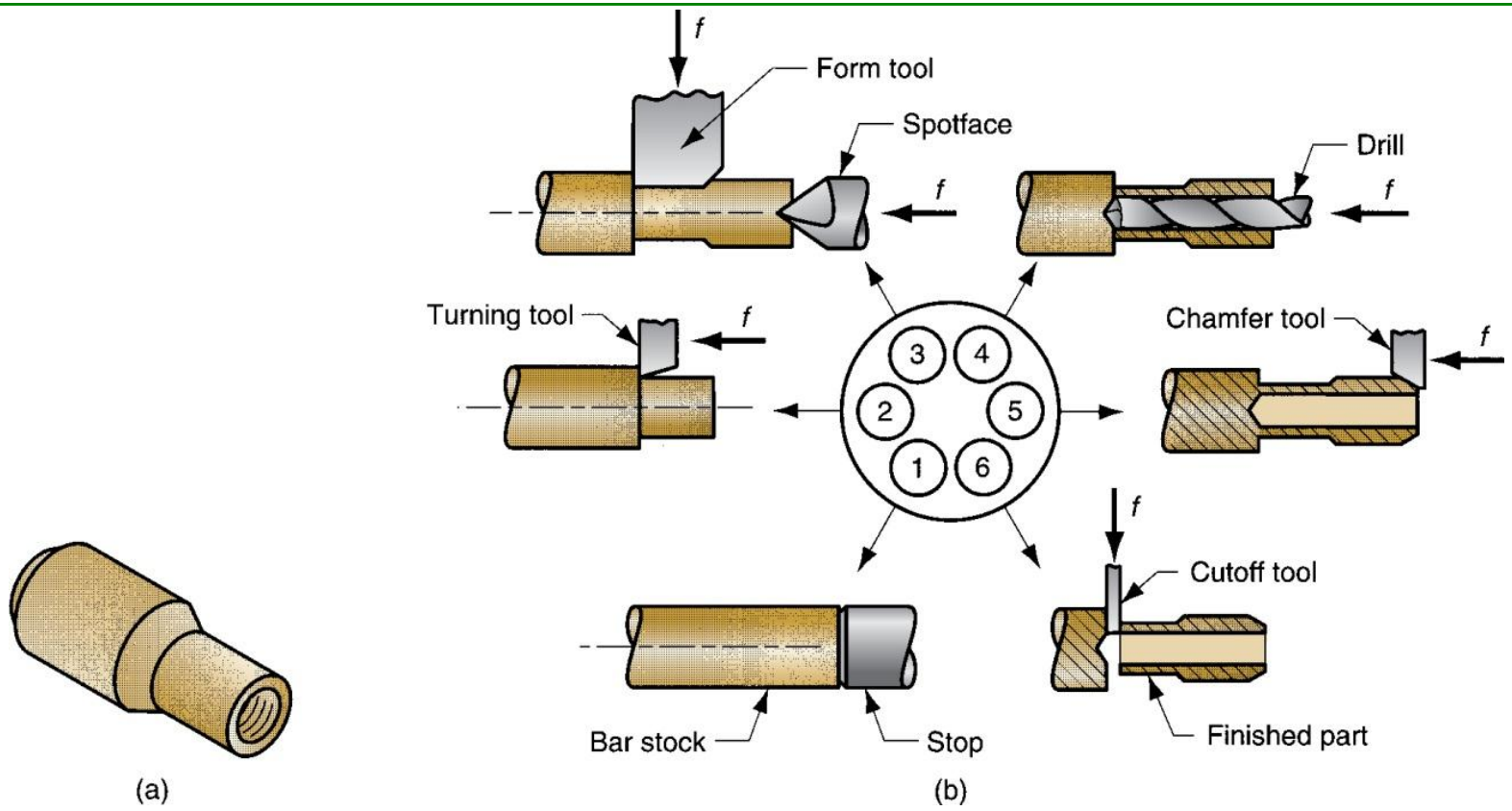
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Multiple Spindle Bar Machines

- More than one spindle, so multiple parts machined simultaneously by multiple tools
  - ✓ Example: six spindle automatic bar machine works on six parts at a time
- After each machining cycle, spindles (including collets and work-bars) are indexed (rotated) to next position

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Six Spindle Bar Machine



(a) Part

(b) sequence of operations: (1) feed stock to stop, (2) turn main diameter, (3) form second diameter and spotface, (4) drill, (5) chamfer, and (6) cutoff

Boring

- **Difference between boring and turning:**
  - ✓ Boring is performed on the inside diameter of an existing hole
  - ✓ Turning is performed on the outside diameter of an existing cylinder. In effect, boring is an internal turning operation

Boring machines

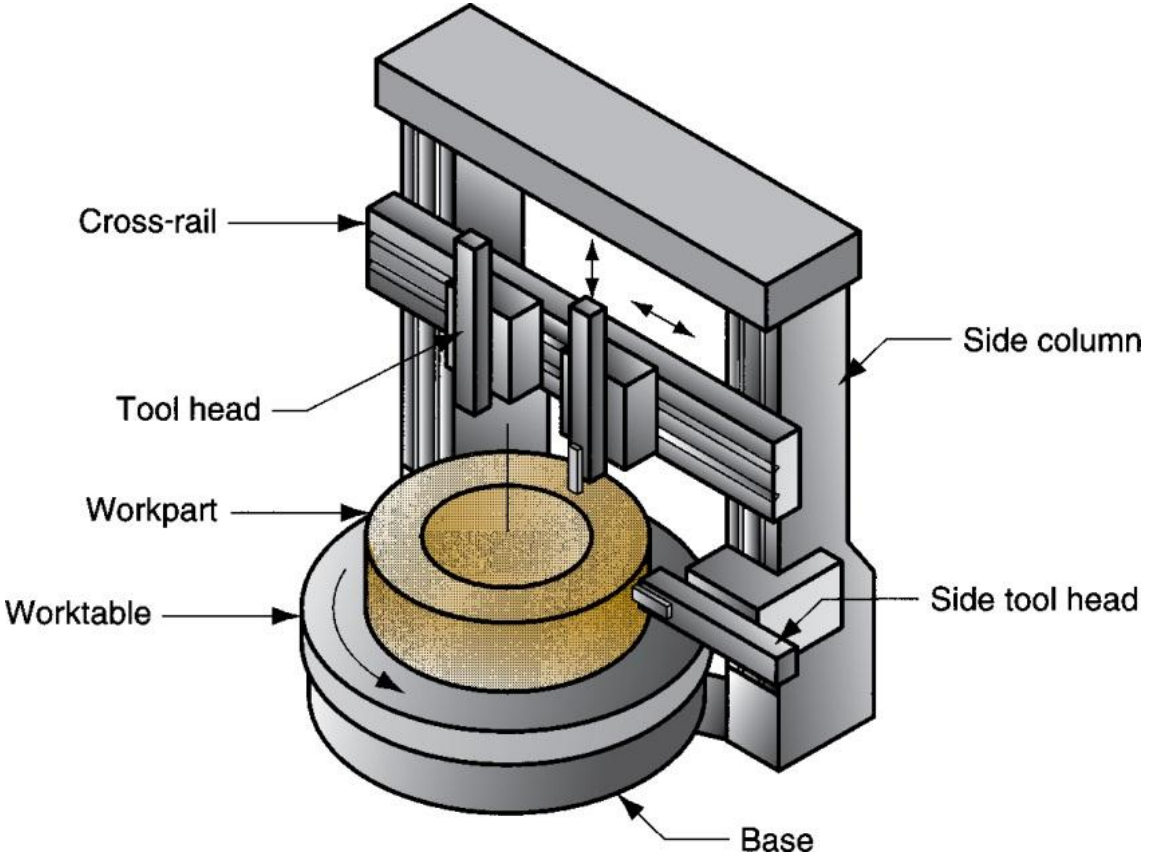
- Horizontal or vertical - refers to the orientation of the axis of rotation of machine spindle

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Vertical Boring Mill

Applications:

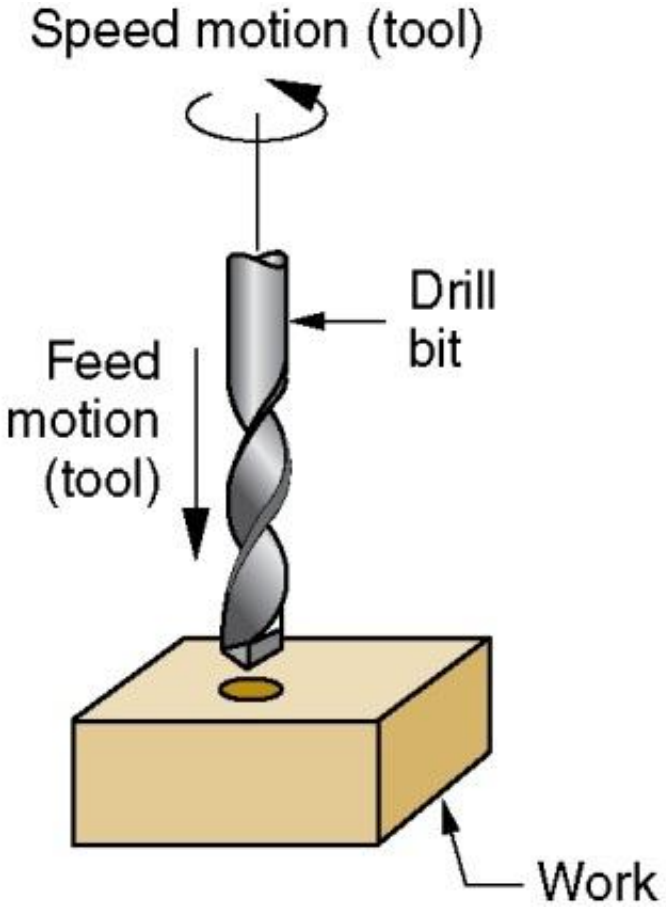
Large, heavy work-parts that have low L/D ratio



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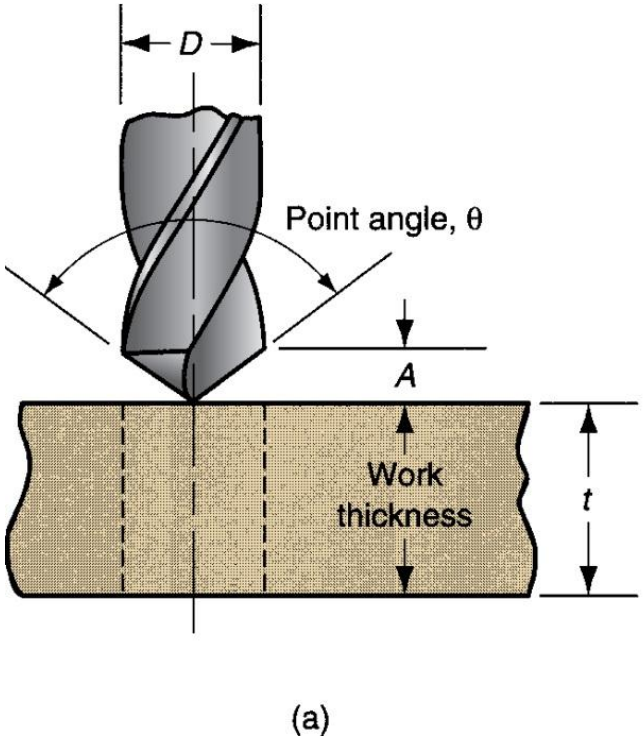
Drilling

- Creates a round hole in a work-part
- Compare to boring which can only enlarge an existing hole
- Cutting tool is called a ***drill*** or ***drill bit.***
- Machine tool: ***drill press***

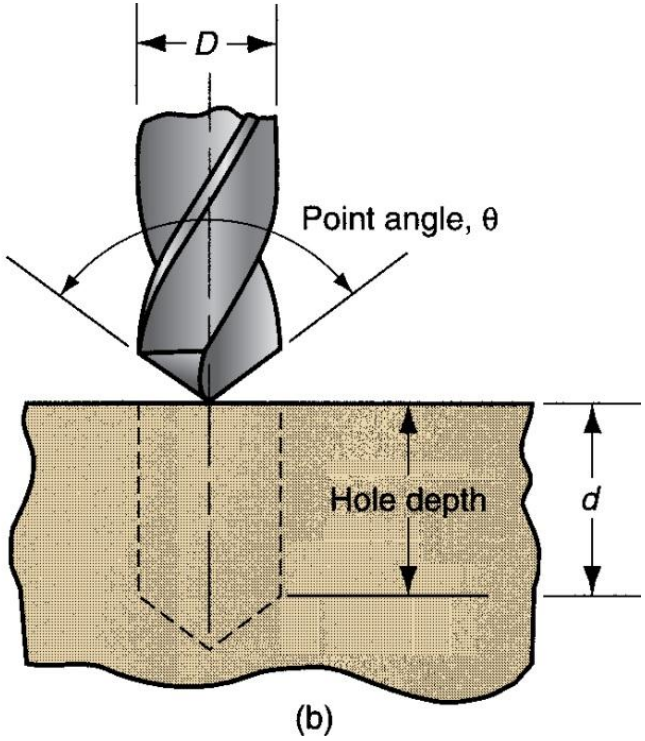


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Through Hole and Blind Hole

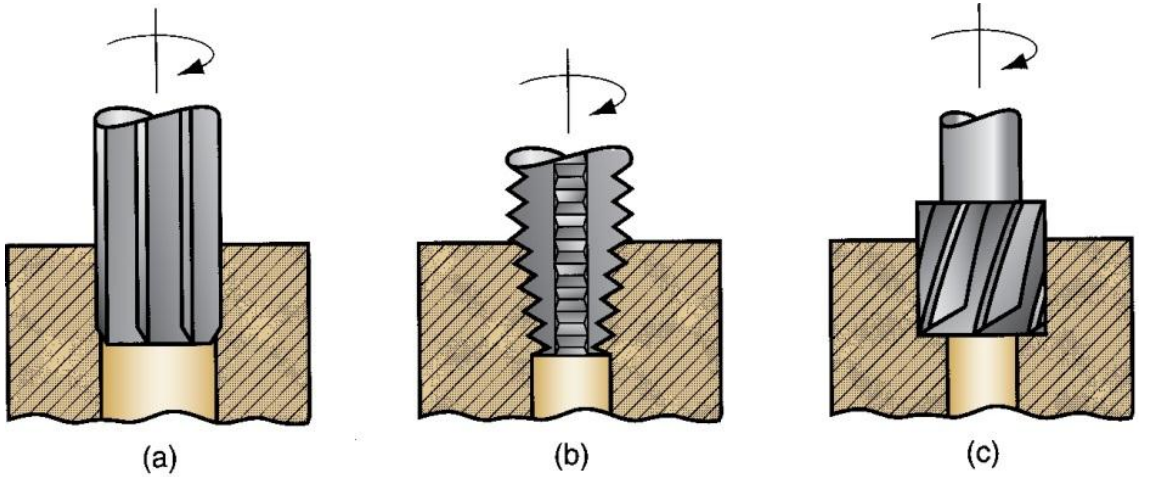


(a) Through hole - drill exits opposite side of work

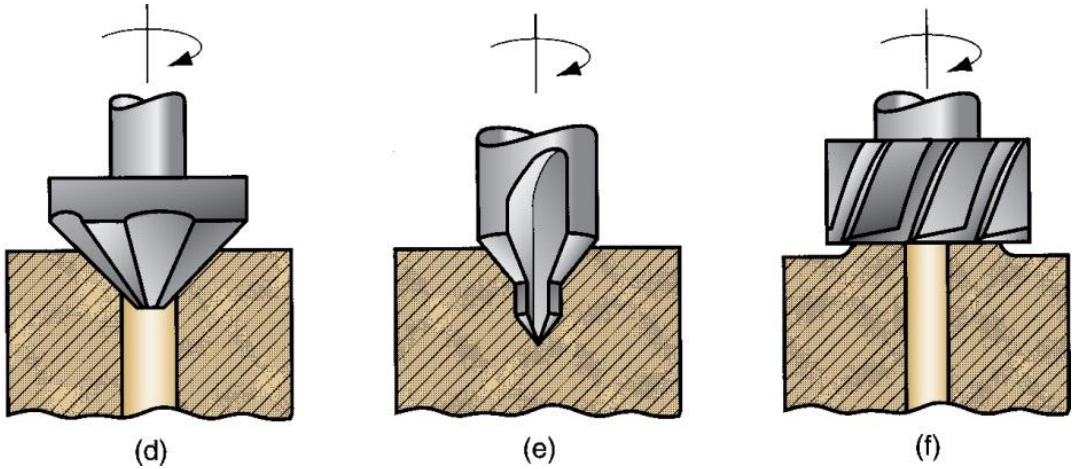


(b) blind hole – drill does not exit opposite side

Drilling Operations



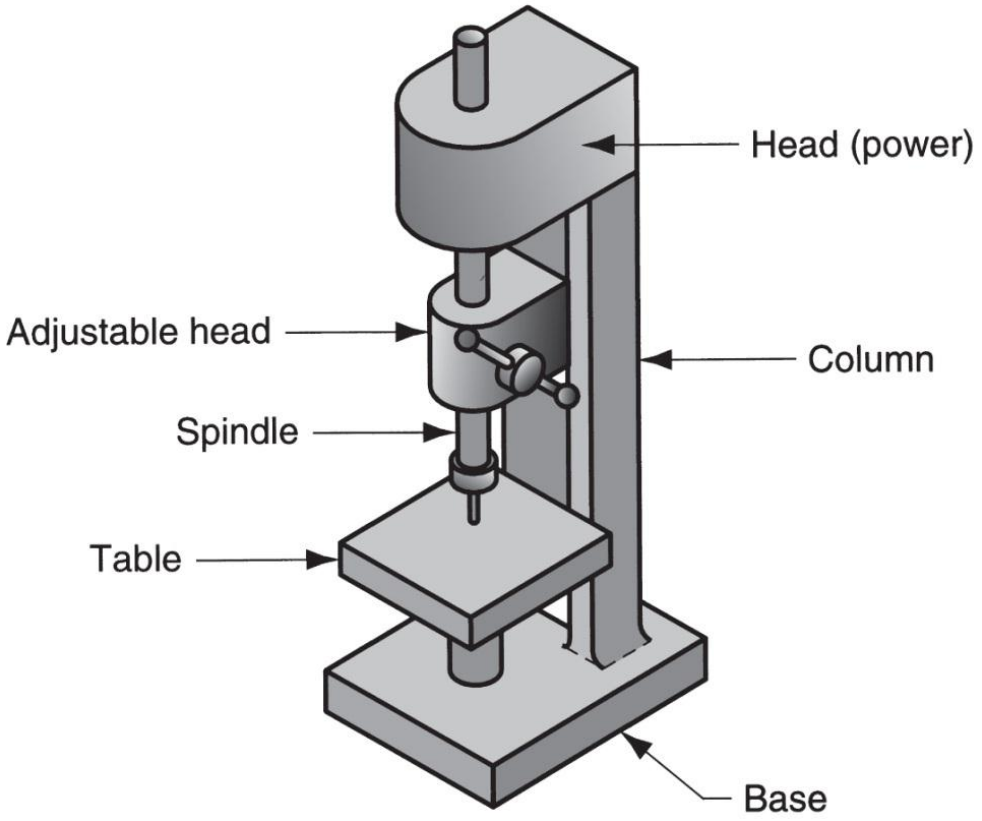
Reaming Tapping Counter-boring



Countersinking Center drilling Spot facing

Drill Press

- Upright drill press that stands on the floor
- Similar to bench drill, however, bench drill is mounted on a table or bench



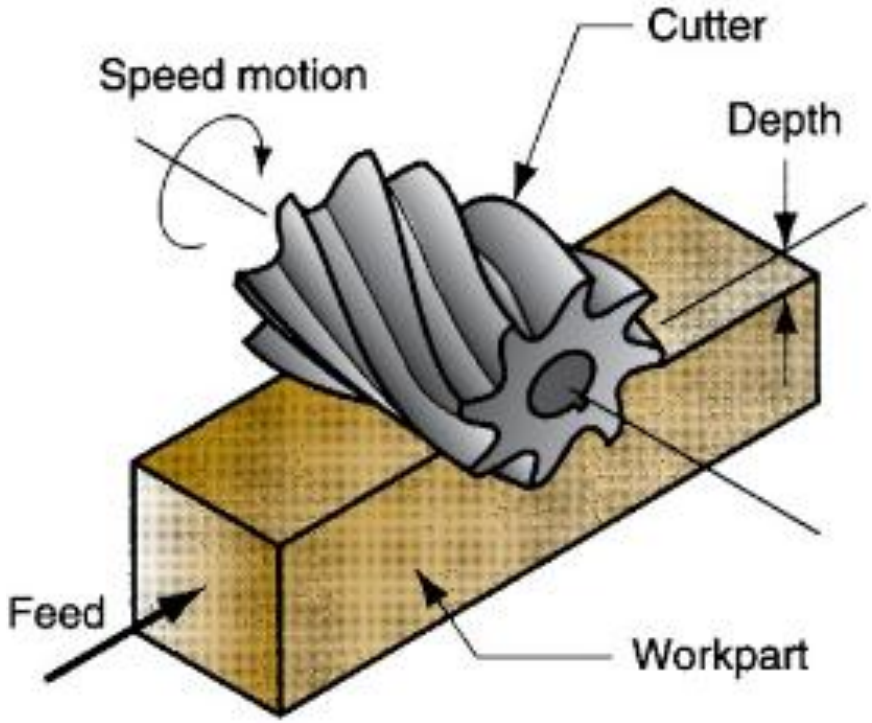
Milling

- Machining operation in which work is fed past a rotating tool with multiple cutting edges.
- Axis of tool rotation is perpendicular to feed.
- Cutting tool is called a milling cutter and cutting edges are called teeth.
- Machine tool is called a milling machine.
- Basic milling operation creates a planar surface, however, other geometries are possible.
- There are two forms of milling: ***peripheral milling*** and ***face milling***.

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Peripheral Milling

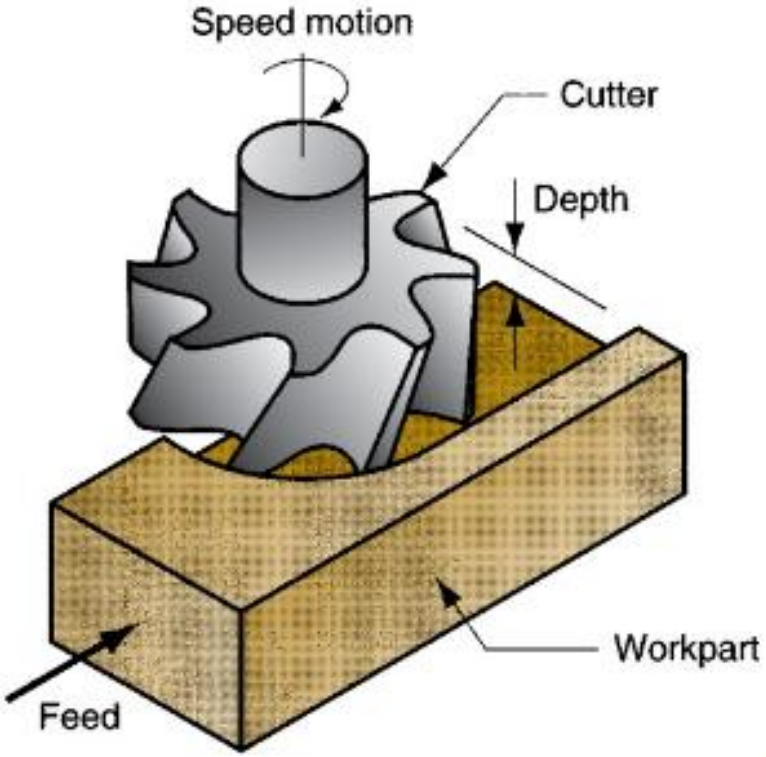
- Cutter axis parallel to surface being machined
- Cutting edges on outside periphery of cutter



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Face Milling

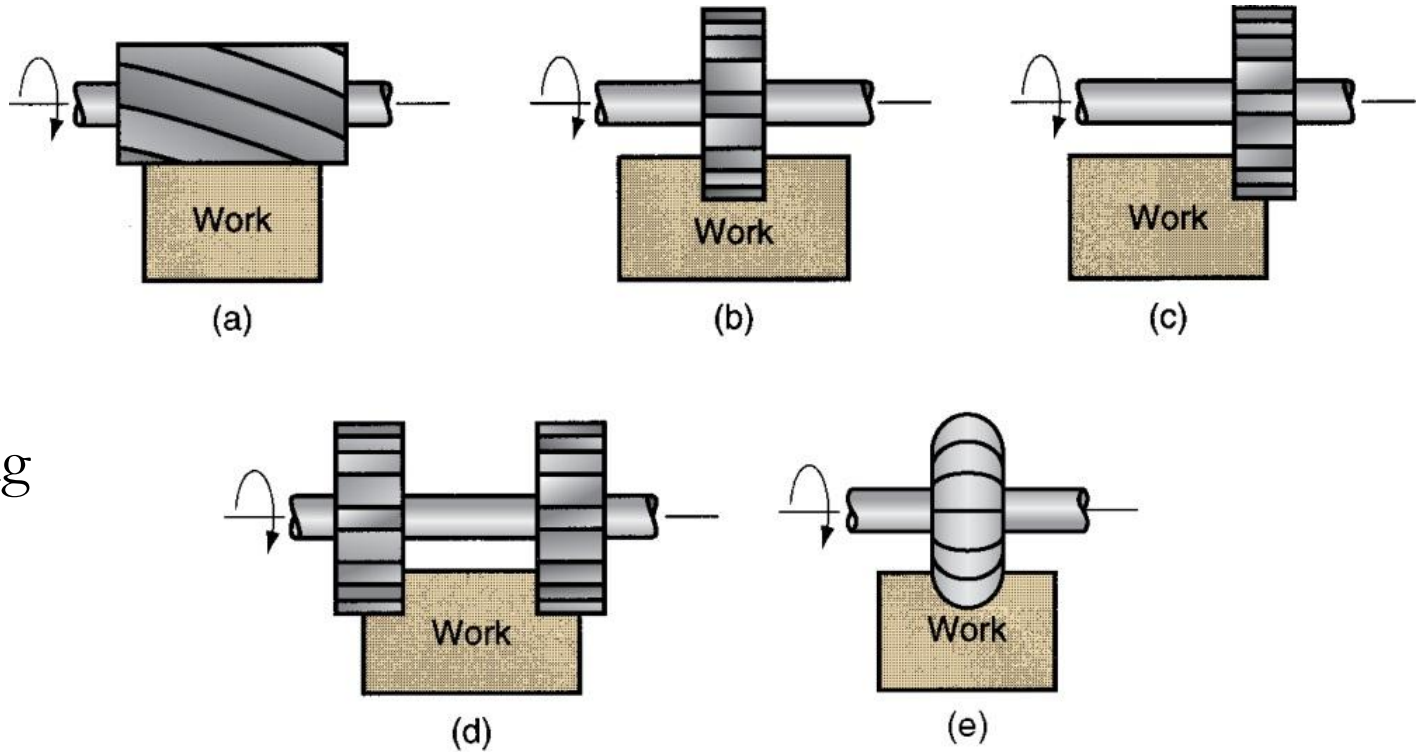
- Cutter axis perpendicular to surface being milled.
- Cutting edges on both the end and outside periphery of the cutter.



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Types of Peripheral Milling

(a) Slab milling
(b) Slotting
(c) Side milling
(d) Straddle milling
(e) form milling



Types of Face Milling

(a) Conventional face milling

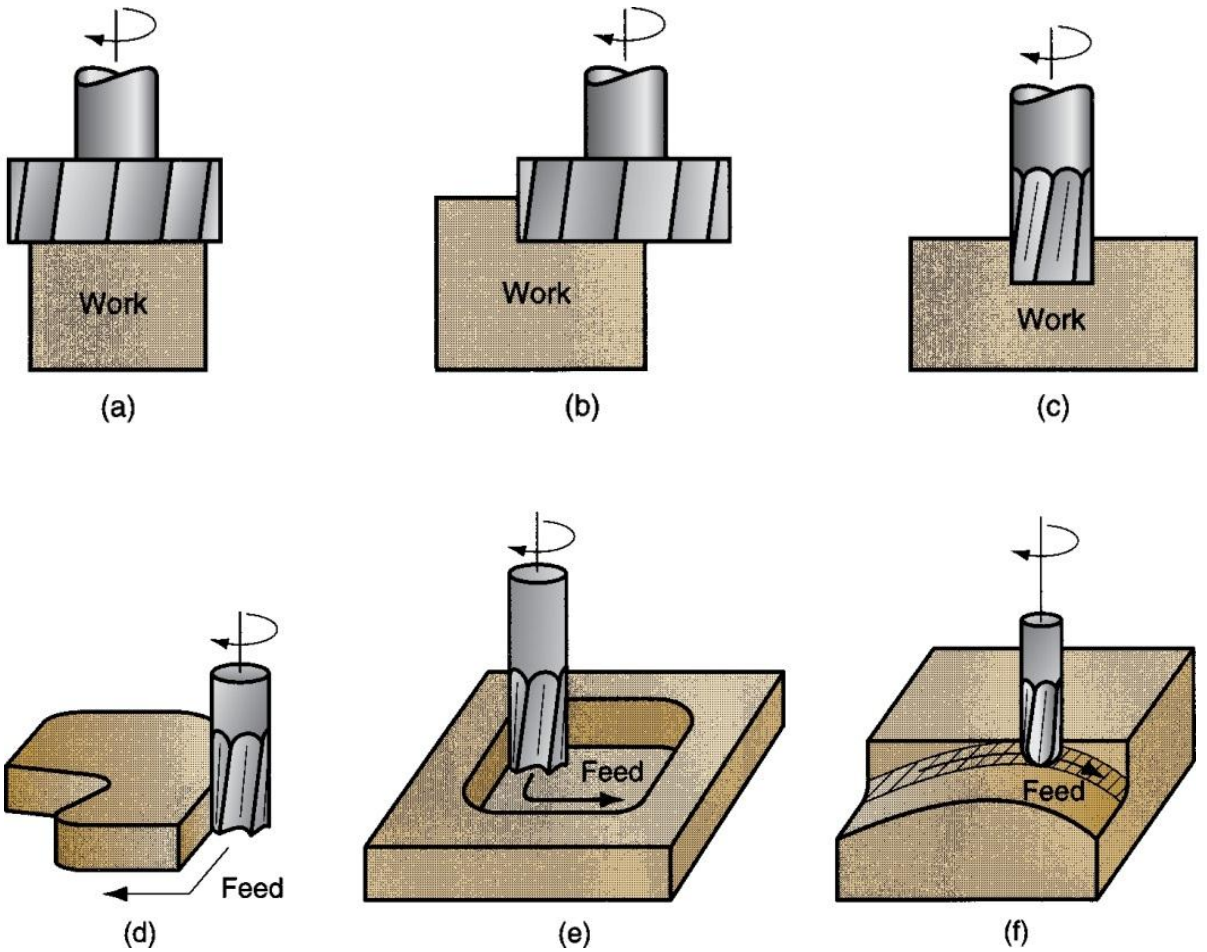
(b) partial face milling

(c) end milling

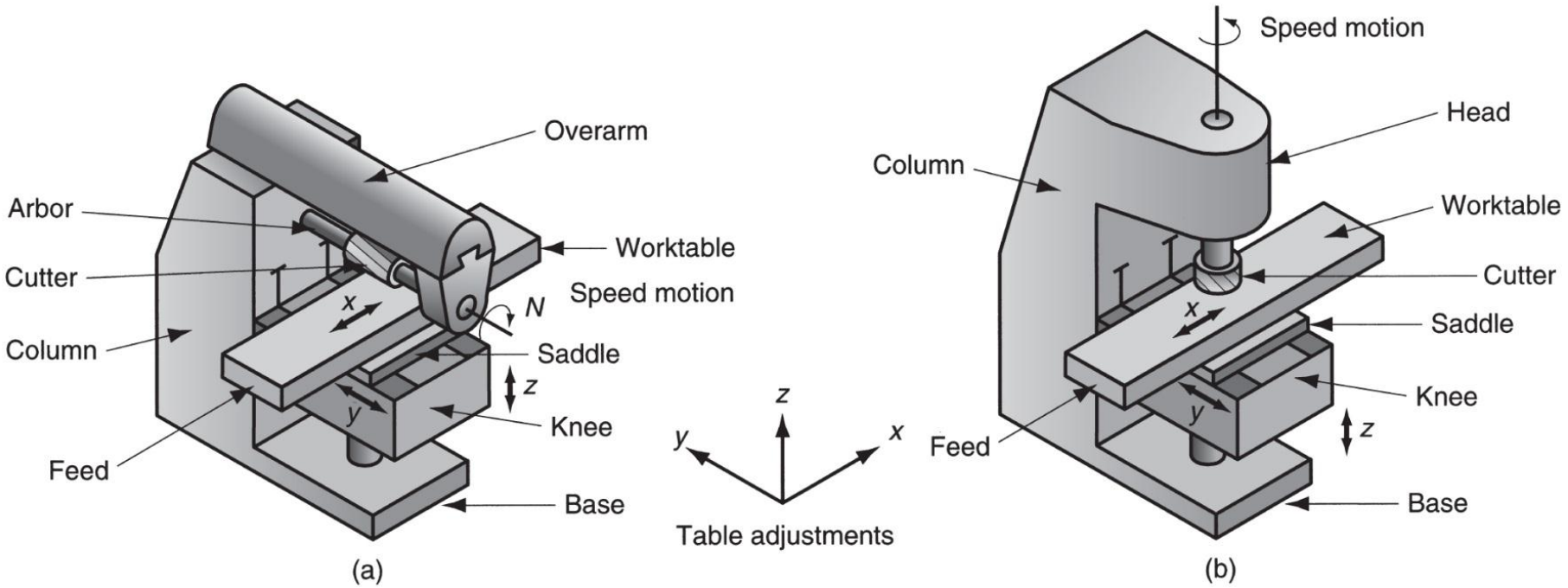
(d) Profile milling

(e) pocket milling

(f) surface contouring



Knee-And-Column Milling Machines



(a) Horizontal and (b) vertical knee-and-column milling machines

Machining Centre

- Highly automated machine tool that can perform multiple machining operations under CNC control in one setup with minimal human attention
- Typical operations are milling and drilling
- Three, four, or five axes
- Other features include automatic tool-changing, Pallet shuttles, automatic work-part positioning

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CNC Machining Centre



Mill-Turn Centers

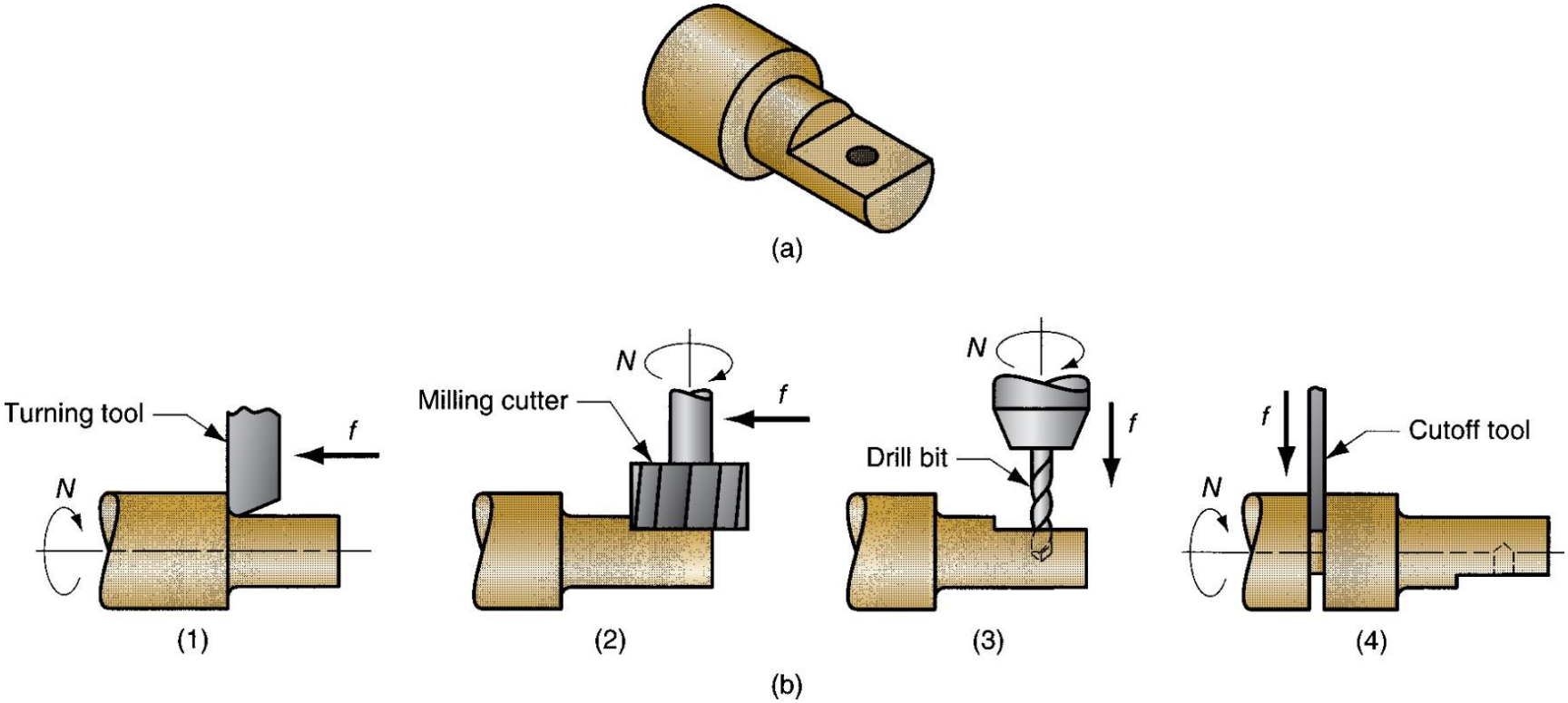
- Highly automated machine tool that can perform turning, milling, and drilling operations in one setup.
- Can position a cylindrical work-part at a specified angle so a rotating cutting tool (e.g., milling cutter) can machine features into outside surface of part.
- Conventional turning center cannot stop work-part at a defined angular position and does not include rotating tool spindles.

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CNC Turning Center - Industrial Robot to Load and Unload Parts



Operation of Mill-Turn Centres

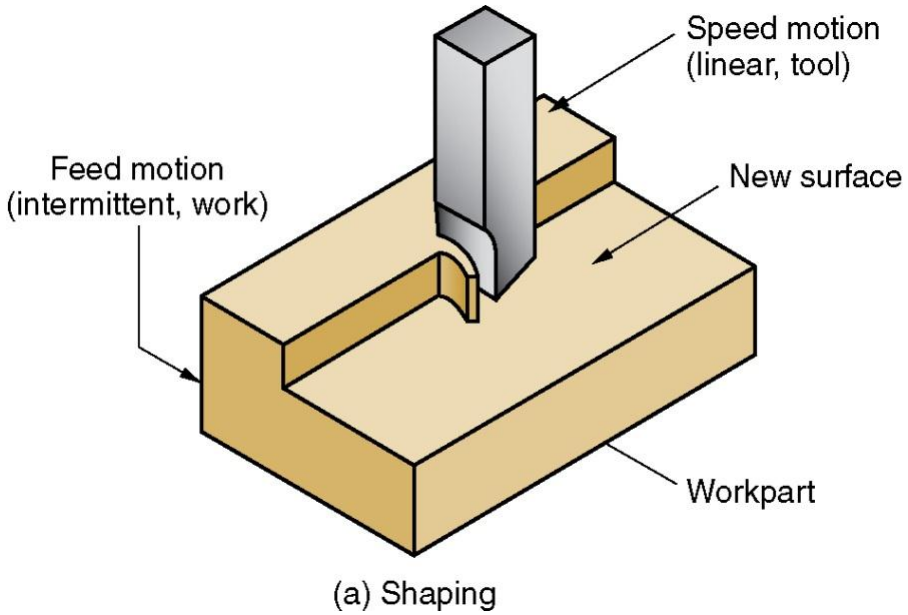


(a) Part

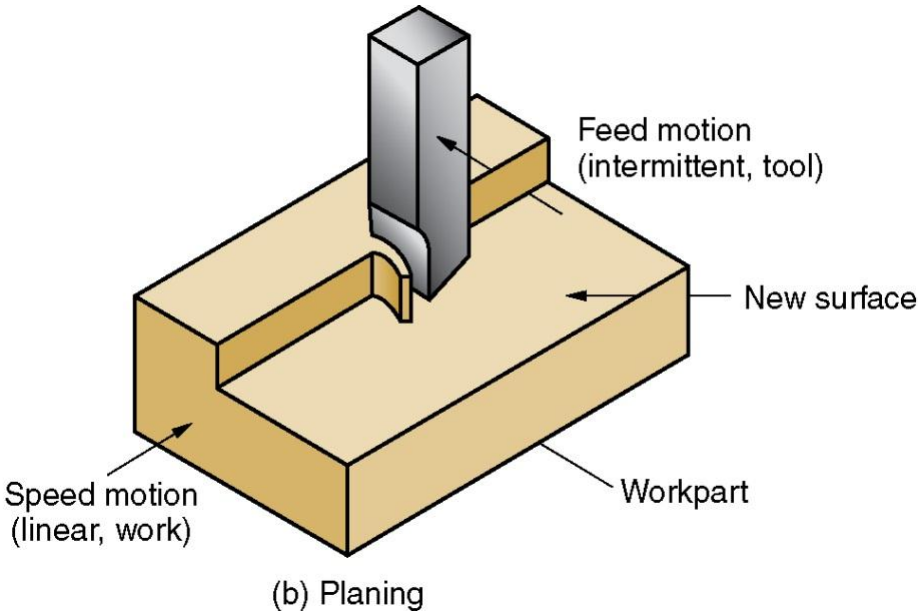
(b) Sequence of operations : (1) turn second diameter, (2) mill flat, (3) drill hole, and (4) cutoff

Shaping and Planing

- Shaping and planning are similar operations.
- Both use a single point cutting tool that moves linearly relative to the work-part.



(a) Shaping



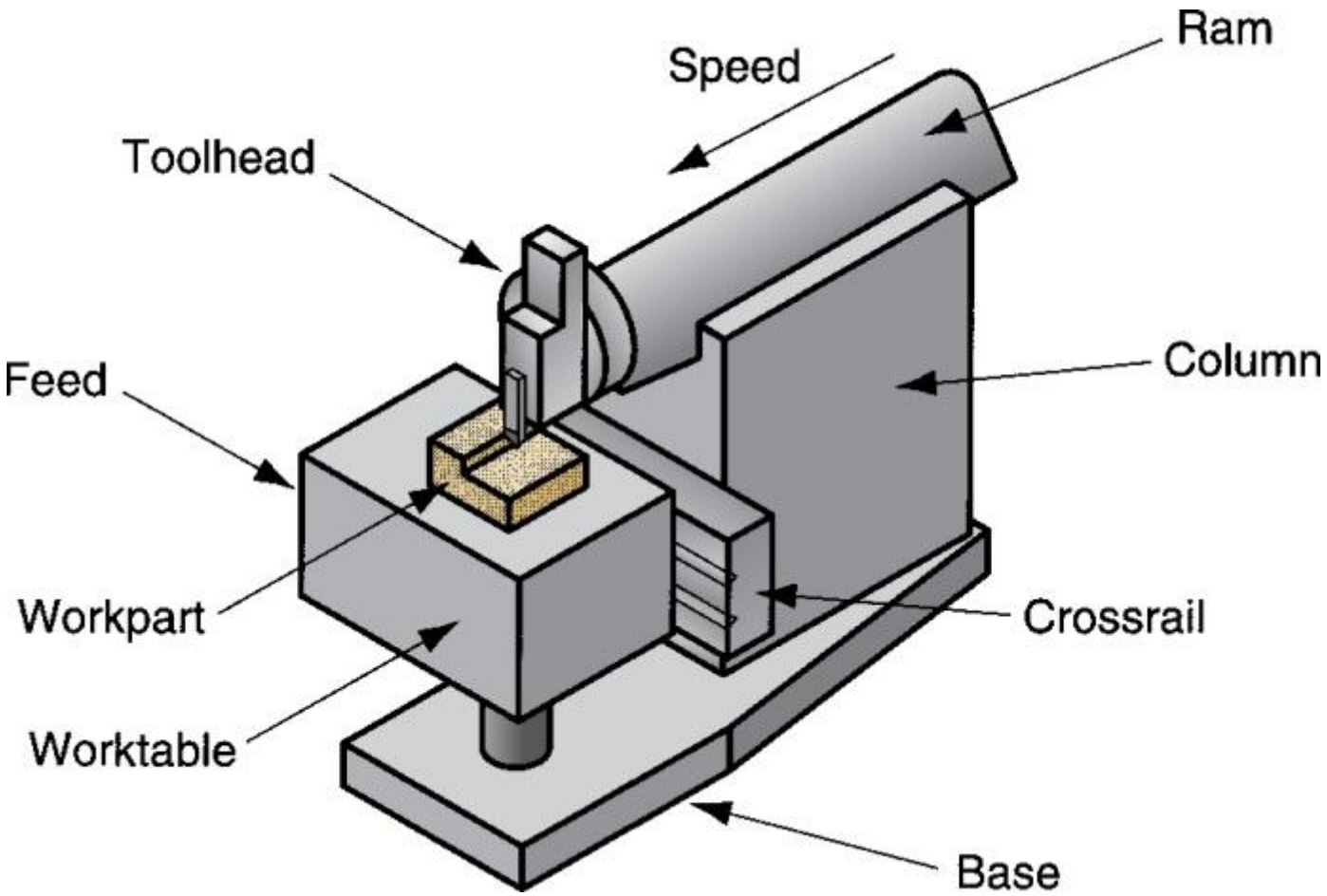
(b) Planing

Shaping and Planing

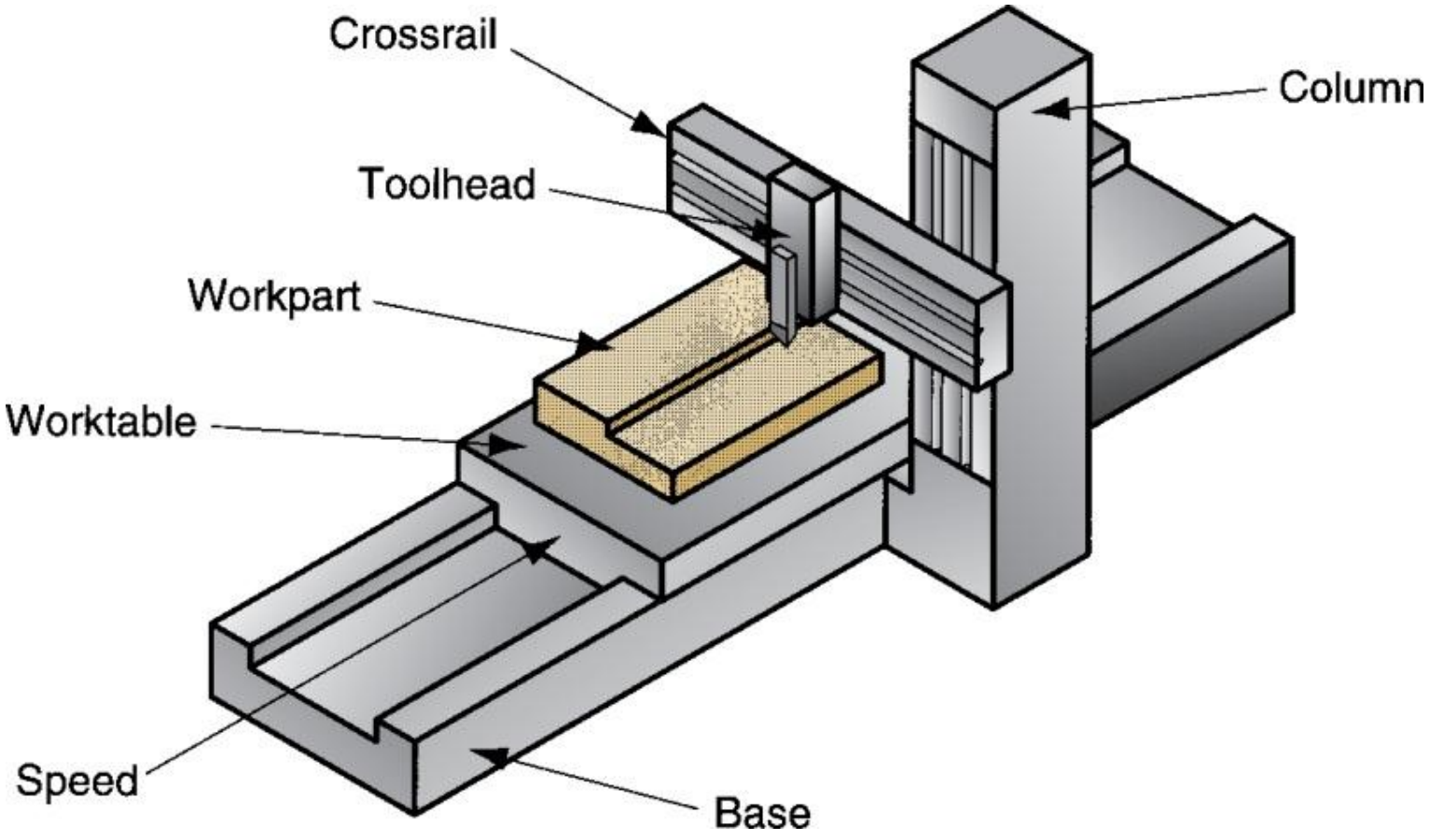
- A straight, flat surface is created in both operations
- They have interrupted cutting operation
- Tools are subjected to impact loading when entering work. Typical tooling include single point high speed steel tools
- Have low cutting speeds due to start-and-stop motion

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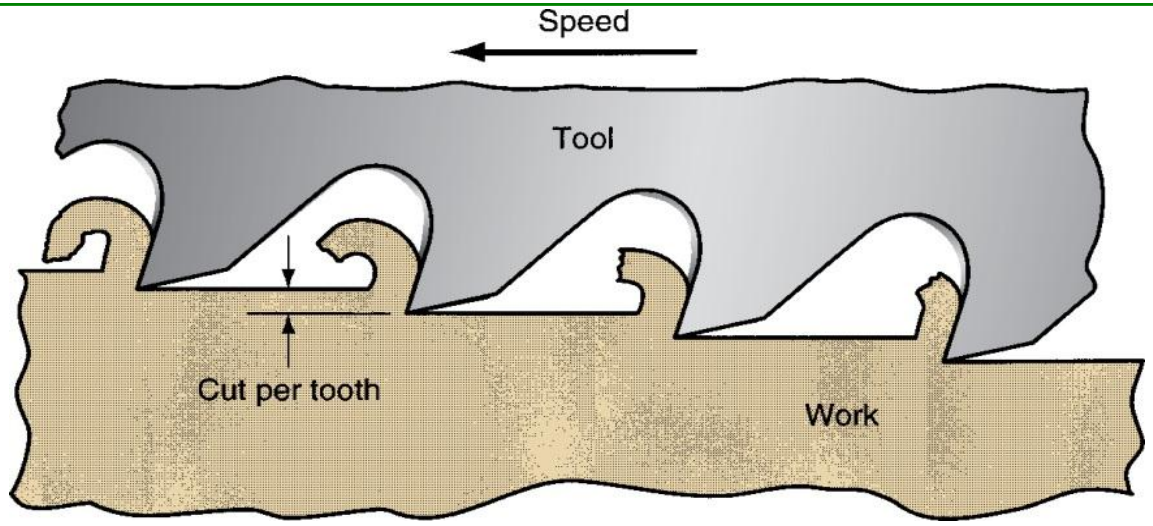
Shaper


Ram
Speed
Toolhead
Feed
Column
Workpart
Crossrail
Worktable
Base

Planer



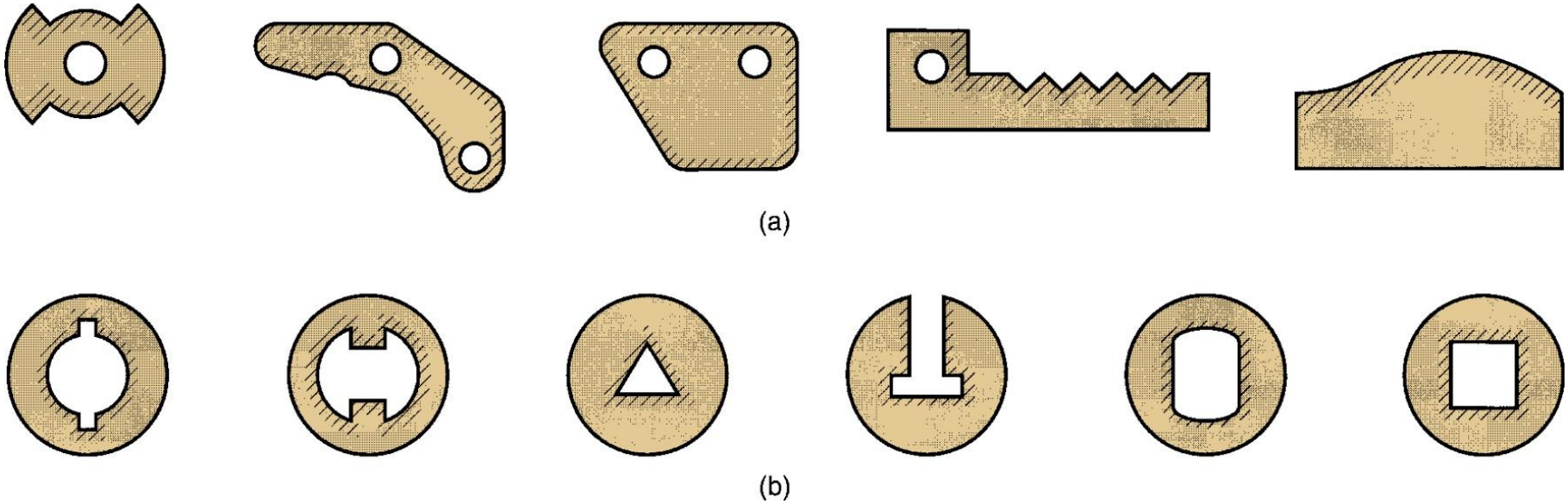
Broaching



- The movement of a multiple tooth cutting tool linearly relative to work in direction of tool axis.
- Cutting tool is called a ***broach.***
- ***Advantages:*** Good surface finish, close tolerances, variety of work shapes possible.
- ***Disadvantages:*** Tooling is expensive.

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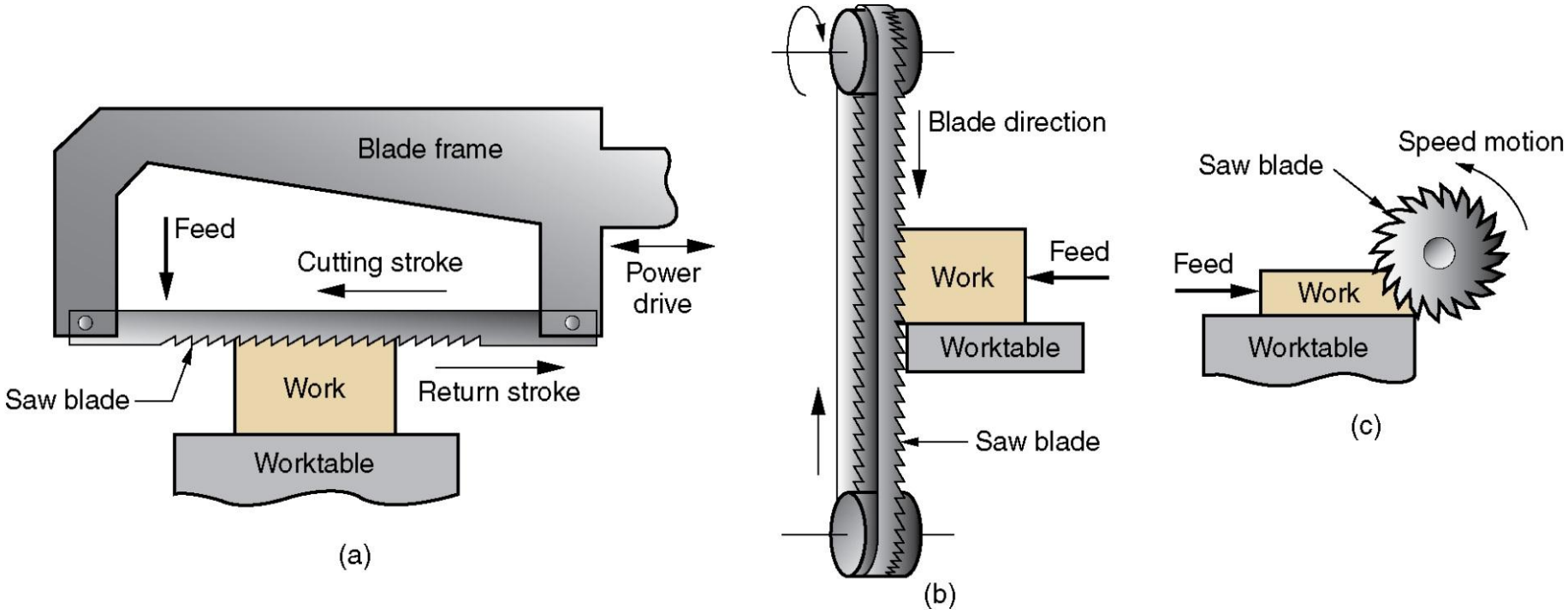
Broaching



(a) External
(b) Internal broaching (cross-hatching indicates surface broached)

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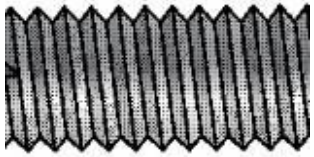
Sawing



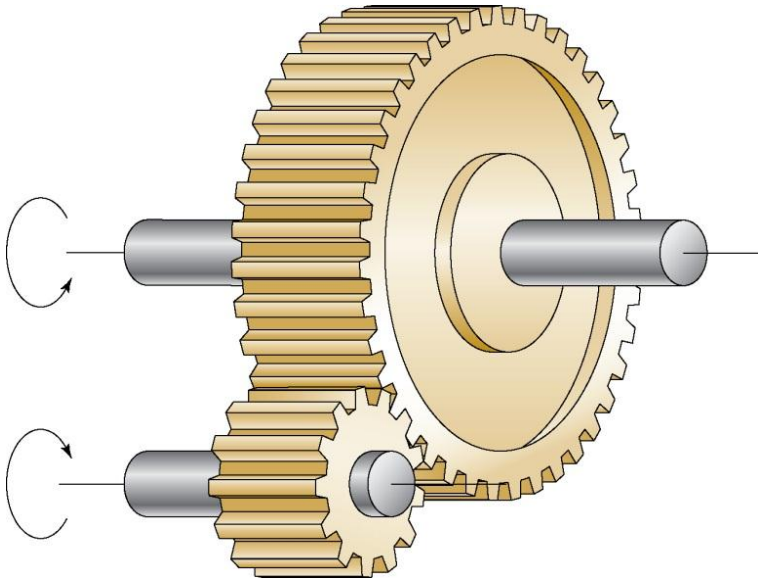
(a) Power hacksaw, (b) bandsaw (vertical), and (c) circular saw

Machining Operations for Special Geometries

- **Screw threads**



- **Gear Teeth**



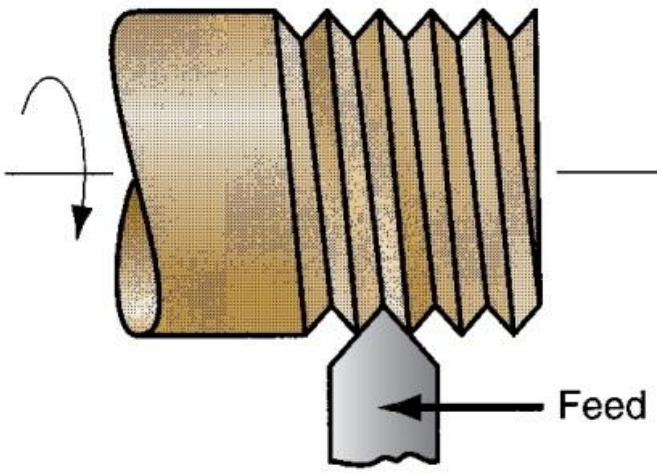
Cutting Screw Threads

- Methods for producing external threads
  - Single-point threading
  - Threading die
  - Thread chasing using self-opening threading dies
  - Thread milling

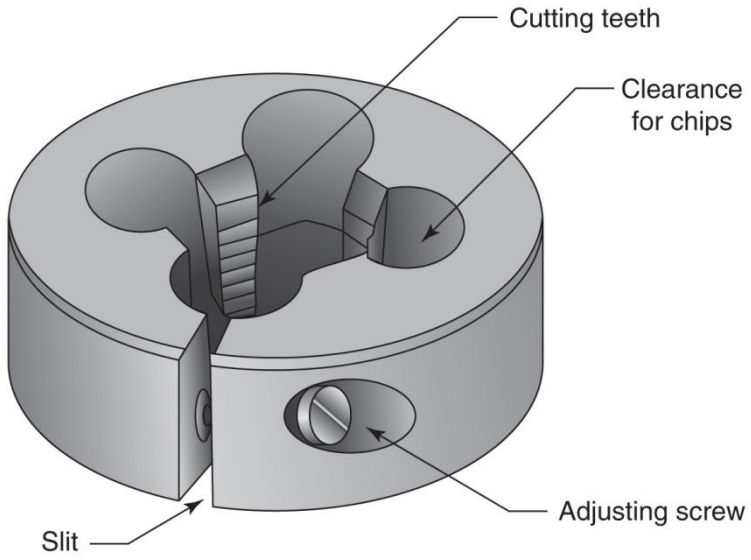
- Methods for producing internal threads
  - Tapping - using a solid tap
  - Collapsible taps - cutting teeth retract for quick removal from hole

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Cutting external screw threads

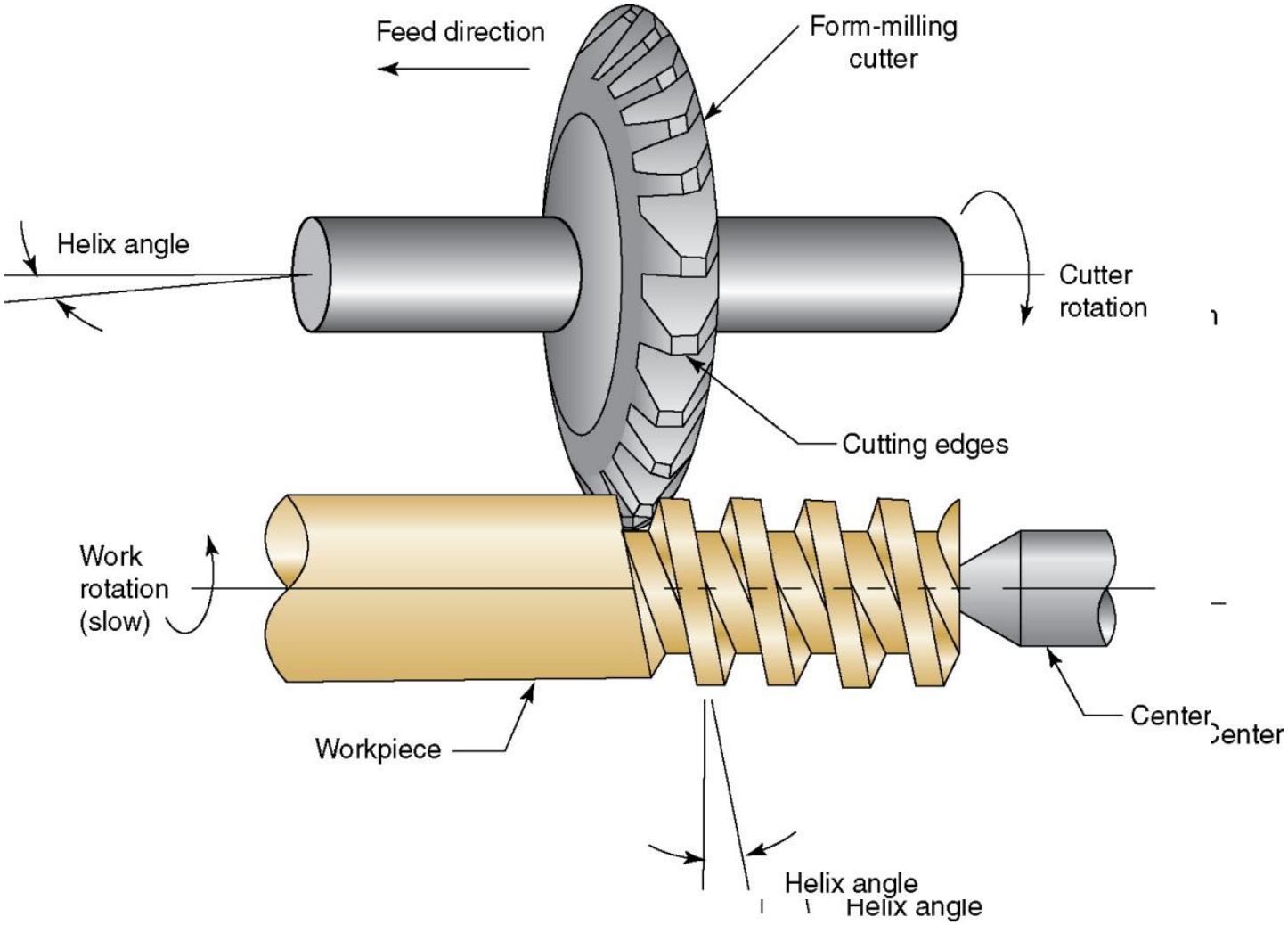


Single-point thread cutting



Threading die

Thread Milling Using a Form-Milling Cutter



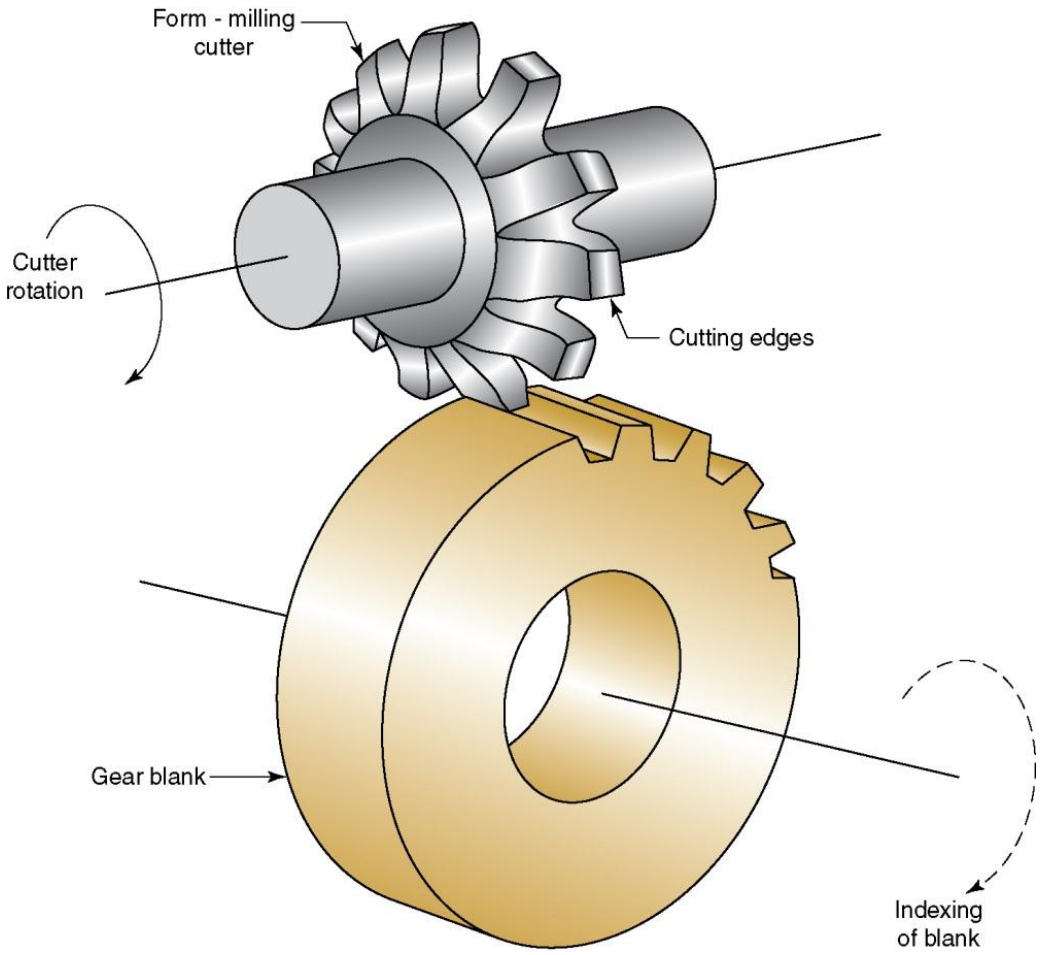
Principal Operations for Machining Gear Teeth

- Form milling - use of a form milling cutter.
- Gear hobbing - also milling but using a special cutter called a ***hob.***
- Gear shaping – There are two forms:
  - Single point tool to gradually shape each gear tooth spacing.
  - Cutter has general shape of the gear but with cutting teeth on one side.
- Gear broaching - for internal and external gears.

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Form Milling of Gear Teeth

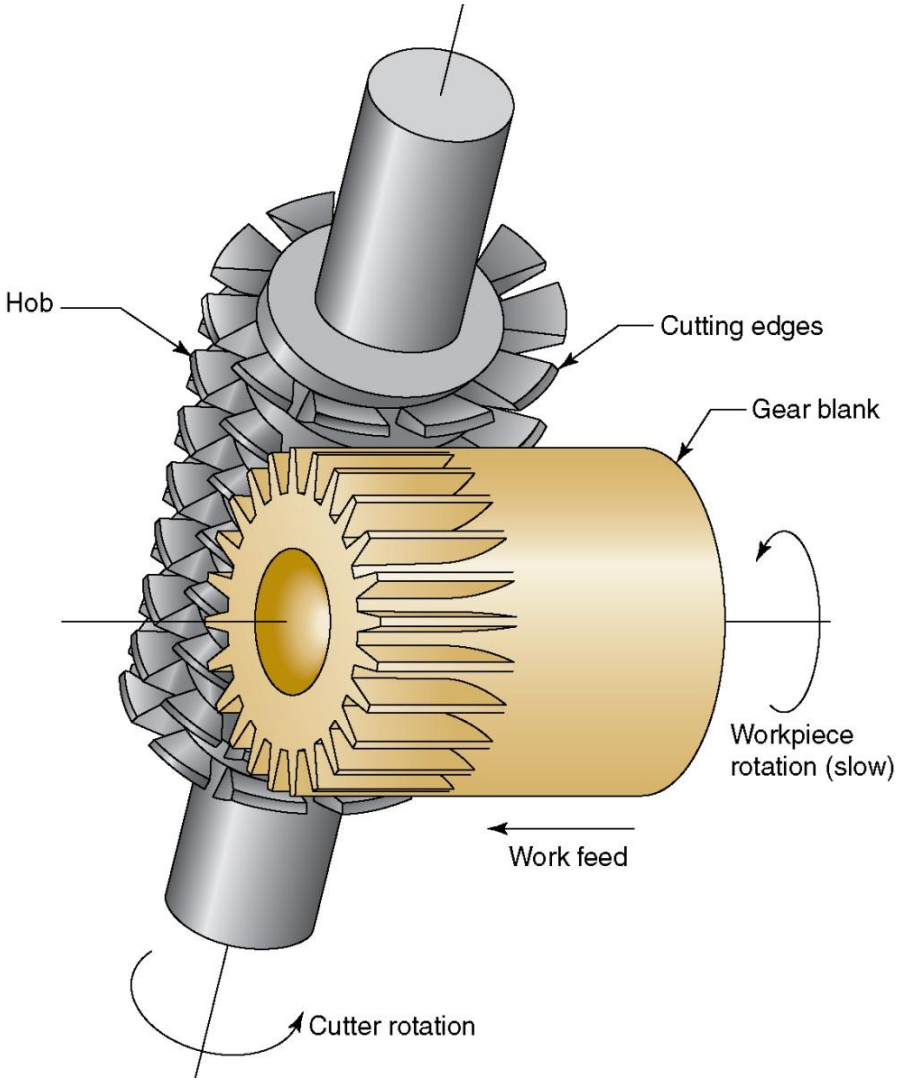
- The form milling cutter has teeth with the shape of the spaces between teeth on the gear.
- Gear blank is indexed between each pass to establish correct size of the gear tooth.



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Gear Hobbing

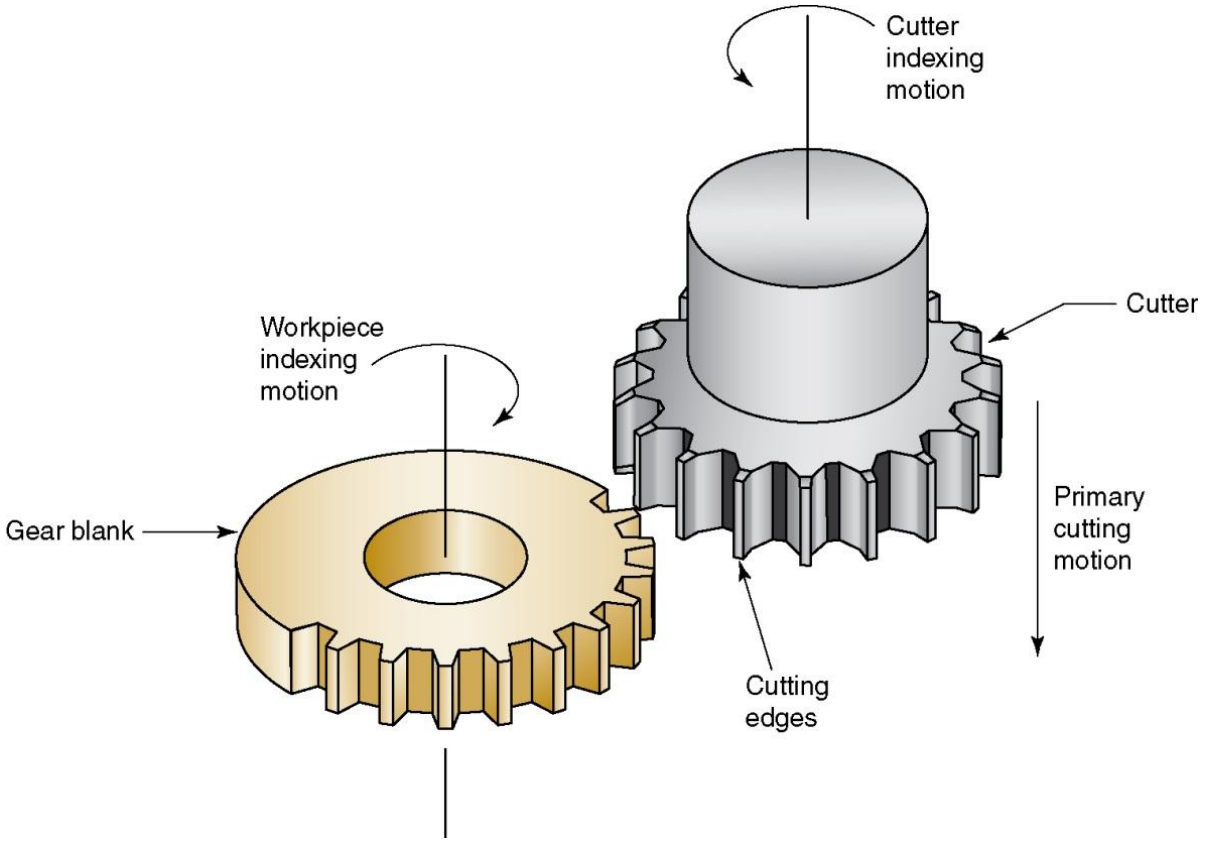
- Hob has a slight helix and its rotation is coordinated with much slower rotation of the gear blank.
- Special milling machines (called ***hobbing machines***) accomplish the relative speed and feed motions between cutter and gear blank.



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Gear Shaping

- To start the process, cutter is gradually fed into gear blank.
- Then, cutter and blank are slowly rotated after each stroke to maintain tooth spacing.
- Performed on special machines called ***gear shapers.***



Gear Broaching

- Applicable for both external gears and internal gears (teeth on inside of gear).
- Cost of tooling (broach) is high due to its complex geometry.
- For internal gears, broach consists of a series of gear-shaped cutting teeth of increasing size to form the gear teeth in successive steps as broach is drawn through starting hole.
- For external gears, broach is tubular with inward-facing cutting teeth.

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THANK YOU